Models_10Sep2025

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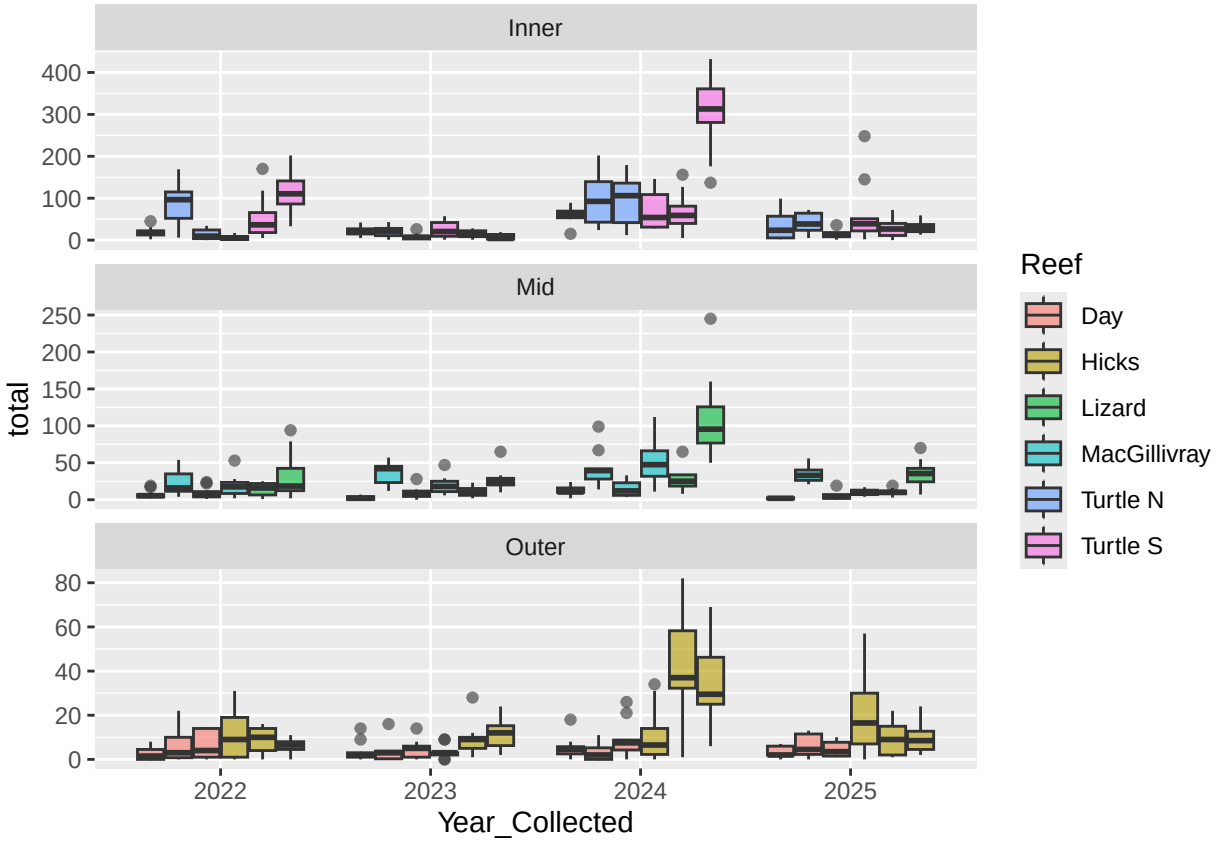
2025-10-29



Summary Statistics

total_tiles	total_recruits	total_acro	total_poci	total_poritidae	total_bottom	total_side	total_top
707	20966	15383	3295	464	11957	7315	1694

Temporal statistics

Year_Collected	total_tiles	total_recruits	total_acro	total_poci	total_poritidae	total_bottom	total_side	total_top
2022	174	4395	3156	634	125	2687	1488	220

Year_Collected	total_tiles	total_recruits	total_acro	total_poci	total_poritidae	total_bottom	total_side	total_top
2023	175	2364	1277	738	60	1763	513	88
2024	179	10604	8439	1211	127	5381	4264	959
2025	179	3603	2511	712	152	2126	1050	427

Run generalised linear models with the Poisson, normal binomail and the zero inflated normal bionomial. Total is the dependent variable (the count of coral recruits at a given sampling unit) Shelf position is a fixed effect predicator, it is used to test if the average coral settlement differs between shelf positions Random effects: a nested random effect, modelling the variations between reefs, and variation between sites nested within reefs

Victor: I recommend including Year as a fixed effect since you are interested in differences in recruitment across years, and the levels are not interchangeable. In contrast, a variable like Site would be more appropriate as a random effect in your analysis.

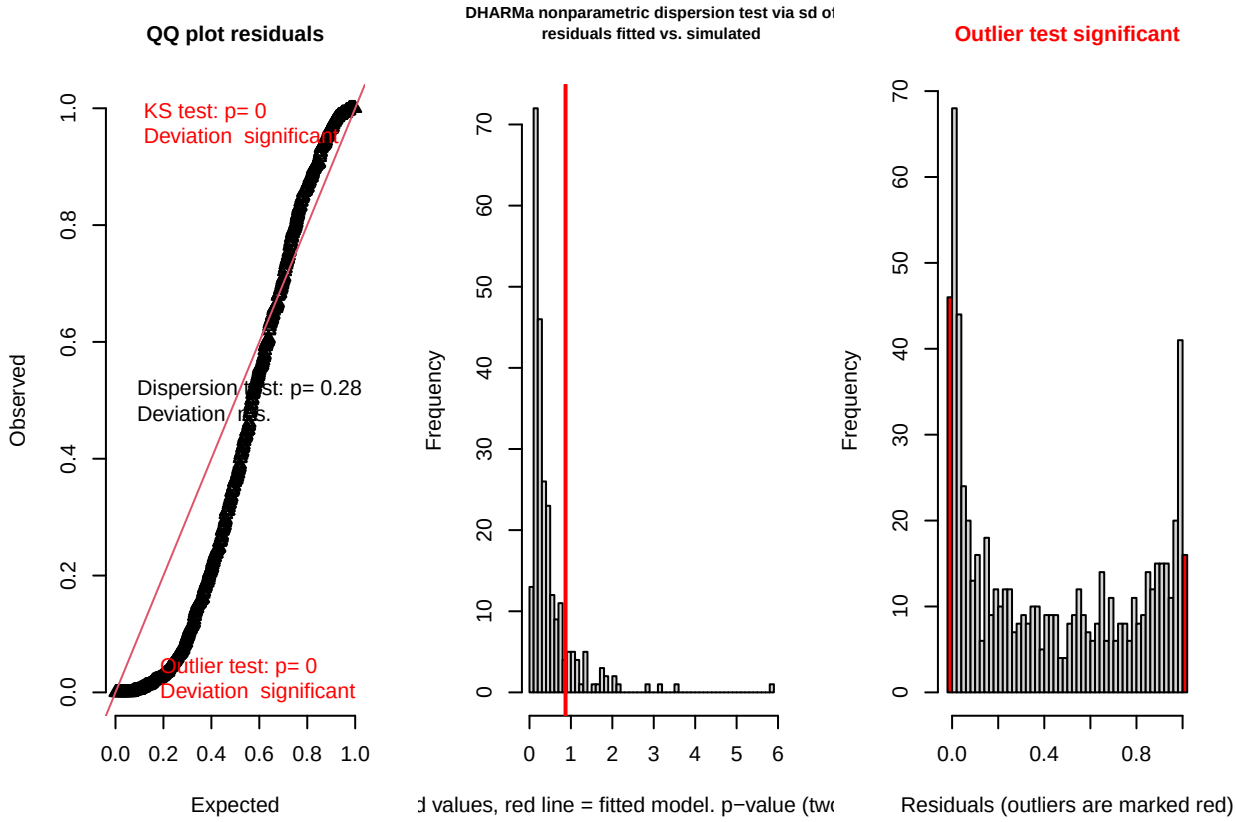
Should I be scaling or centring the data?

```
pois_total_yr <- glmmTMB(total ~ Shelf_Position + Year_Collected + (1 | Reef/Site),
      family = "poisson", data = Coral_settlement_nGBR)

nbin_total_yr <- glmmTMB(total ~ Shelf_Position +  Year_Collected +(1 | Reef/Site),
        family = "nbinom2", data = Coral_settlement_nGBR)

zbin_total_yr <- glmmTMB(total ~ Shelf_Position + Year_Collected + (1 | Reef/Site),
                      ziformula = ~1,
        family = "nbinom2", data = Coral_settlement_nGBR)

pois_total_yr2 <- glmmTMB(total ~ Shelf_Position * Year_Collected + (1 | Reef/Site),
      family = "poisson", data = Coral_settlement_nGBR)

nbin_total_yr2 <- glmmTMB(total ~ Shelf_Position * Year_Collected +(1 | Reef/Site),
        family = "nbinom2", data = Coral_settlement_nGBR)

zbin_total_yr2 <- glmmTMB(total ~ Shelf_Position* Year_Collected + (1 | Reef/Site),
                      ziformula = ~1,
        family = "nbinom2", data = Coral_settlement_nGBR)
```

Use DHARMA to assess residuals

```
sim1.1 <- simulateResiduals(pois_total_yr)
testResiduals(sim1.1) #not good fit
```

```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```

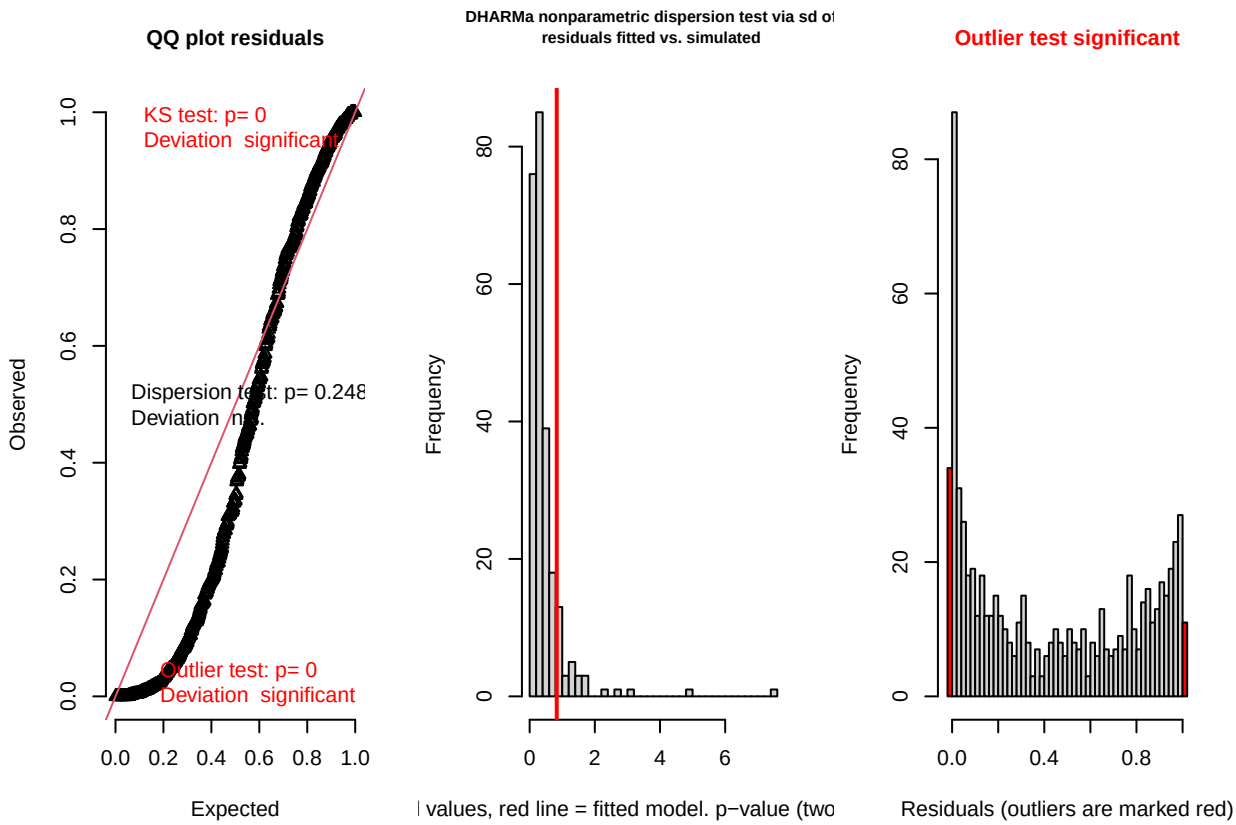



```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.2092, p-value < 2.2e-16
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 1.7732, p-value = 0.28
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 82, observations = 707, p-value < 2.2e-16
```

```
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.09331522 0.14191105
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                 0.115983
```

```
sim1.2 <- simulateResiduals(pois_total_yr2)
testResiduals(sim1.2) #not good fit
```

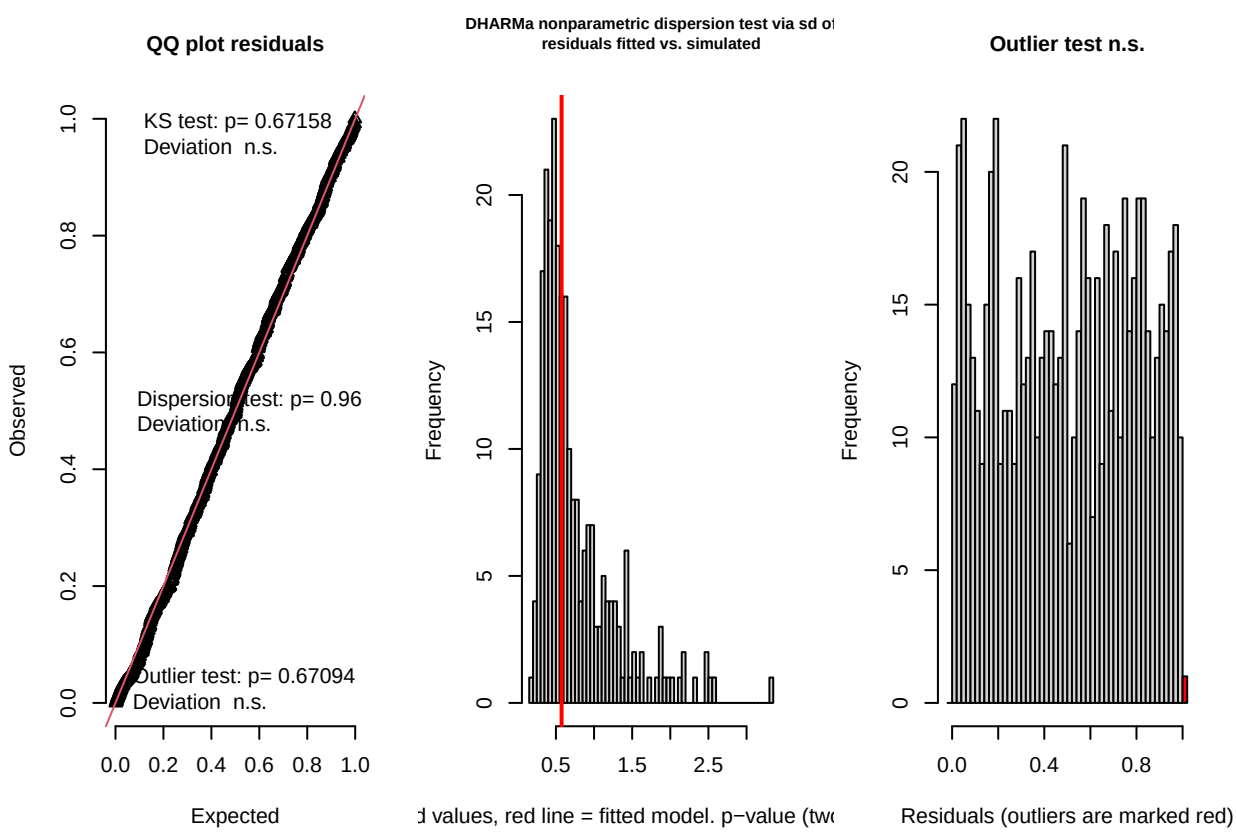
```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.21028, p-value < 2.2e-16
## alternative hypothesis: two-sided
##
##
## $dispersion
##
```

```
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 1.7891, p-value = 0.248
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 66, observations = 707, p-value < 2.2e-16
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.07293411 0.11723537
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.09335219
```

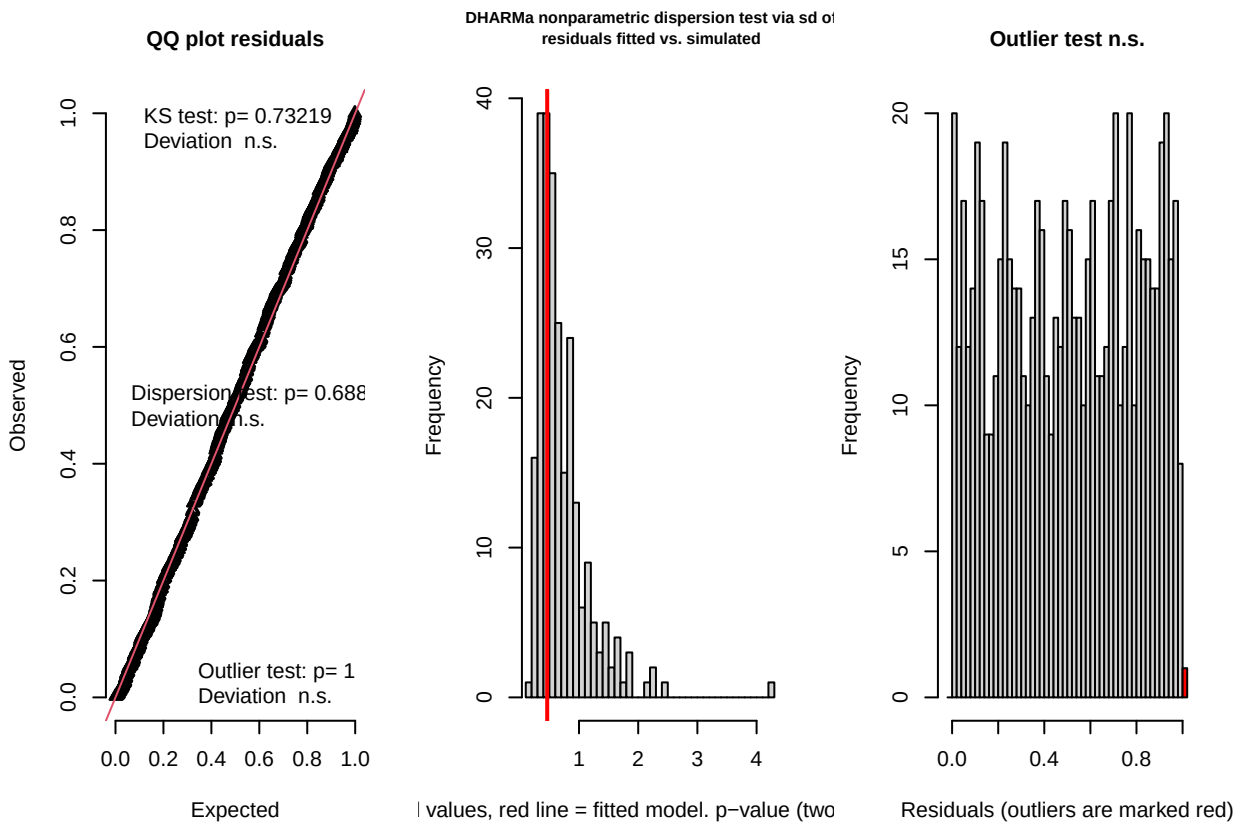
```
sim1.3 <- simulateResiduals(nbin_total_yr)
testResiduals(sim1.3) #good
```




```
## $uniformity
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.027216, p-value = 0.6716
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.74701, p-value = 0.96
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 4, observations = 707, p-value = 0.6709
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.001543622 0.014422145
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                          0.005657709
```

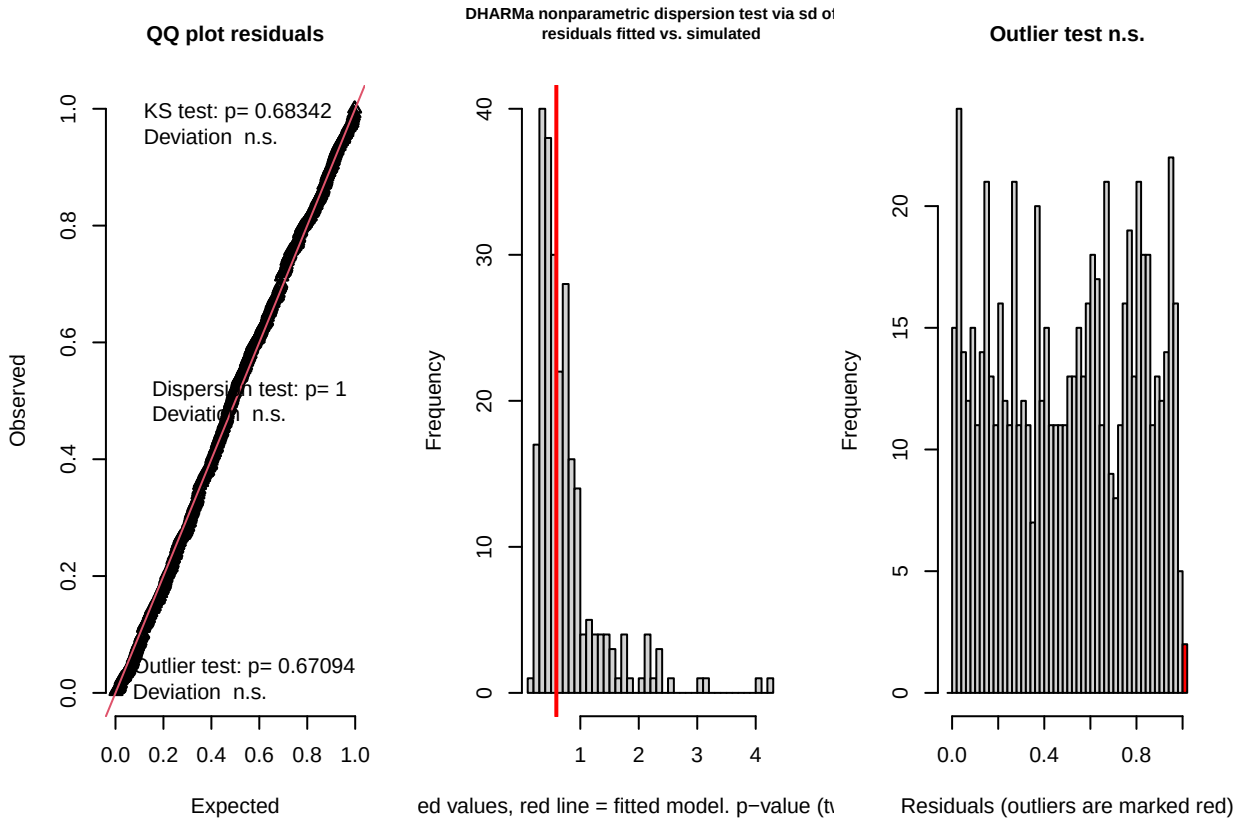
```r
sim1.4 <- simulateResiduals(nbin_total_yr2)
testResiduals(sim1.4) #good
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.02585, p-value = 0.7322
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.64497, p-value = 0.688
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 5, observations = 707, p-value = 1
```

```
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.002300174 0.016426295
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.007072136
```

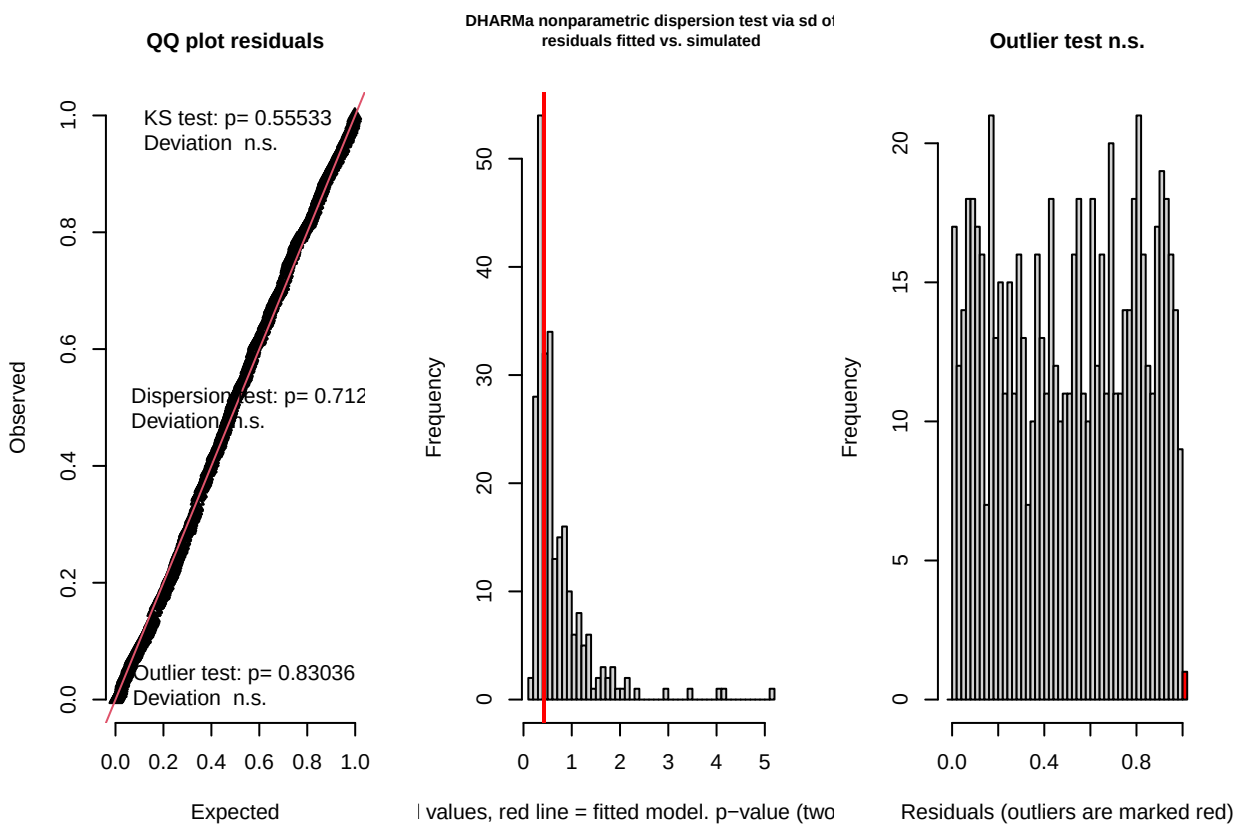
```
sim1.5 <- simulateResiduals(zbin_total_yr)
testResiduals(sim1.5) #good
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.026951, p-value = 0.6834
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
```

```
## dispersion = 0.76656, p-value = 1
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 4, observations = 707, p-value = 0.6709
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.001543622 0.014422145
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.005657709
```

```r
sim1.6 <- simulateResiduals(zbin_total_yr2)
testResiduals(sim1.6) #good
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
```

```
## data:  simulationOutput$scaledResiduals
## D = 0.02983, p-value = 0.5553
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.58511, p-value = 0.712
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 6, observations = 707, p-value = 0.8304
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.003120604 0.018379583
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                            0.008486563
```

Poisson model residuals were bad, therefore will only compare the four models that testResiduals were good for.

```r
AIC(nbin_total_yr, zbin_total_yr, nbin_total_yr2, zbin_total_yr2) |>  #nbin has the lowest AIC with int
  knitr::kable()
```

	df	AIC
nbin_total_yr	9	5620.032
zbin_total_yr	10	5621.740
nbin_total_yr2	15	5600.125
zbin_total_yr2	16	5601.095

negative binomial model has the lowest AIC so I will get the summary of that model.

```
##  Family: nbinom2  ( log )
## Formula:          total ~ Shelf_Position * Year_Collected + (1 | Reef/Site)
## Data: Coral_settlement_nGBR
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    5600.1    5668.5   -2785.1    5570.1       692
##
## Random effects:
```

```
##
## Conditional model:
##  Groups    Name        Variance  Std.Dev.
##  Site:Reef (Intercept) 3.119e-01 0.5584794
##  Reef      (Intercept) 1.946e-07 0.0004412
## Number of obs: 707, groups:  Site:Reef, 18; Reef, 6
##
## Dispersion parameter for nbinom2 family (): 1.36
##
## Conditional model:
##                                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)                              3.7289     0.2565  14.538  < 2e-16 ***
## Shelf_PositionMid                       -0.9463     0.3635  -2.603  0.00923 **
## Shelf_PositionOuter                     -1.8534     0.3669  -5.052 4.38e-07 ***
## Year_Collected2023                      -0.9131     0.1701  -5.368 7.95e-08 ***
## Year_Collected2024                       0.9438     0.1627   5.800 6.64e-09 ***
## Year_Collected2025                      -0.1826     0.1681  -1.087  0.27718
## Shelf_PositionMid:Year_Collected2023     0.7614     0.2406   3.165  0.00155 **
## Shelf_PositionOuter:Year_Collected2023   0.7836     0.2511   3.120  0.00181 **
## Shelf_PositionMid:Year_Collected2024    -0.1874     0.2323  -0.807  0.41977
## Shelf_PositionOuter:Year_Collected2024  -0.2289     0.2415  -0.948  0.34307
## Shelf_PositionMid:Year_Collected2025    -0.1658     0.2388  -0.694  0.48751
## Shelf_PositionOuter:Year_Collected2025   0.4002     0.2454   1.631  0.10291
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2023 and 2024 are significantly different to 2022. 2025 is not significantly different to 2022. Outer shelf is significantly different to inner shelf ($p<0.01$), Mid shelf is significantly different to inner shelf reefs ($p<0.05$). Intercept is also significant: not quite sure what that means in a biological framework.

When making a figure should I then be using summary statistics or raw count data? Because the large number of zeros will make it difficult to do a boxplot of the raw data.

Q2 is the composition of the recruits different across the shelf?

Isolating Data for a single Family of coral

GLMM with Total Recruits for a given coral family, per tile.

To analyze whether there's a difference in the number of recruits of a given coral family (e.g., Poritids) across shelf positions, while accounting for random effects due to site nested within reef and controlling for year, the model you choose depends on:

Whether the response is aggregated or split by tile facet (side/top/bottom),

The structure of your random effects (nested site within reef),

Whether Year_Collected should be a fixed or random effect.

```r
pois_Poritid_fixed <- glmmTMB(Poritid ~ Shelf_Position + Year_Collected+(1 | Reef/Site) ,
      family = "poisson", data = Poritid)

pois_Poritid_interaction <- glmmTMB(Poritid ~ Shelf_Position*Year_Collected + (1 | Reef/Site),
      family = "poisson", data = Poritid)
```

```r
pois_Acroporidae_fixed <- glmmTMB(Acroporidae ~ Shelf_Position + Year_Collected+ (1 | Reef/Site),
      family = "poisson", data = Acroporidae)

pois_Acroporidae_interaction <- glmmTMB(Acroporidae ~ Shelf_Position*Year_Collected + (1 | Reef/Site),
      family = "poisson", data = Acroporidae)

pois_Pocilloporidae_fixed <- glmmTMB(Pocilloporidae ~ Shelf_Position + Year_Collected  + (1 | Reef/Site
      family = "poisson", data = Pocilloporidae)

pois_Pocilloporidae_interaction <- glmmTMB(Pocilloporidae ~ Shelf_Position*Year_Collected + (1 | Reef/S
      family = "poisson", data = Pocilloporidae)
```

Negative Binomial of Total Recruits (for a family)

```r
## Fixed
nbin_Poritid_fixed <- glmmTMB(Poritid ~ Shelf_Position + (1 | Reef/Site) + Year_Collected,
        family = "nbinom2", data = Poritid)

nbin_Pocilloporidae_fixed <- glmmTMB(Pocilloporidae ~ Shelf_Position + (1 | Reef/Site) + Year_Collected
        family = "nbinom2", data = Pocilloporidae)

nbin_Acroporidae_fixed <- glmmTMB(Acroporidae ~ Shelf_Position + (1 | Reef/Site) + Year_Collected,
        family = "nbinom2", data = Acroporidae)

## Interaction
nbin_Poritid_interaction <- glmmTMB(Poritid ~ Shelf_Position*Year_Collected + (1 | Reef/Site),
        family = "nbinom2", data = Poritid)

nbin_Pocilloporidae_interaction <- glmmTMB(Pocilloporidae ~ Shelf_Position * Year_Collected + (1 | Reef
        family = "nbinom2", data = Pocilloporidae)

nbin_Acroporidae_interaction <- glmmTMB(Acroporidae ~ Shelf_Position*Year_Collected + (1 | Reef/Site),
        family = "nbinom2", data = Acroporidae)
```

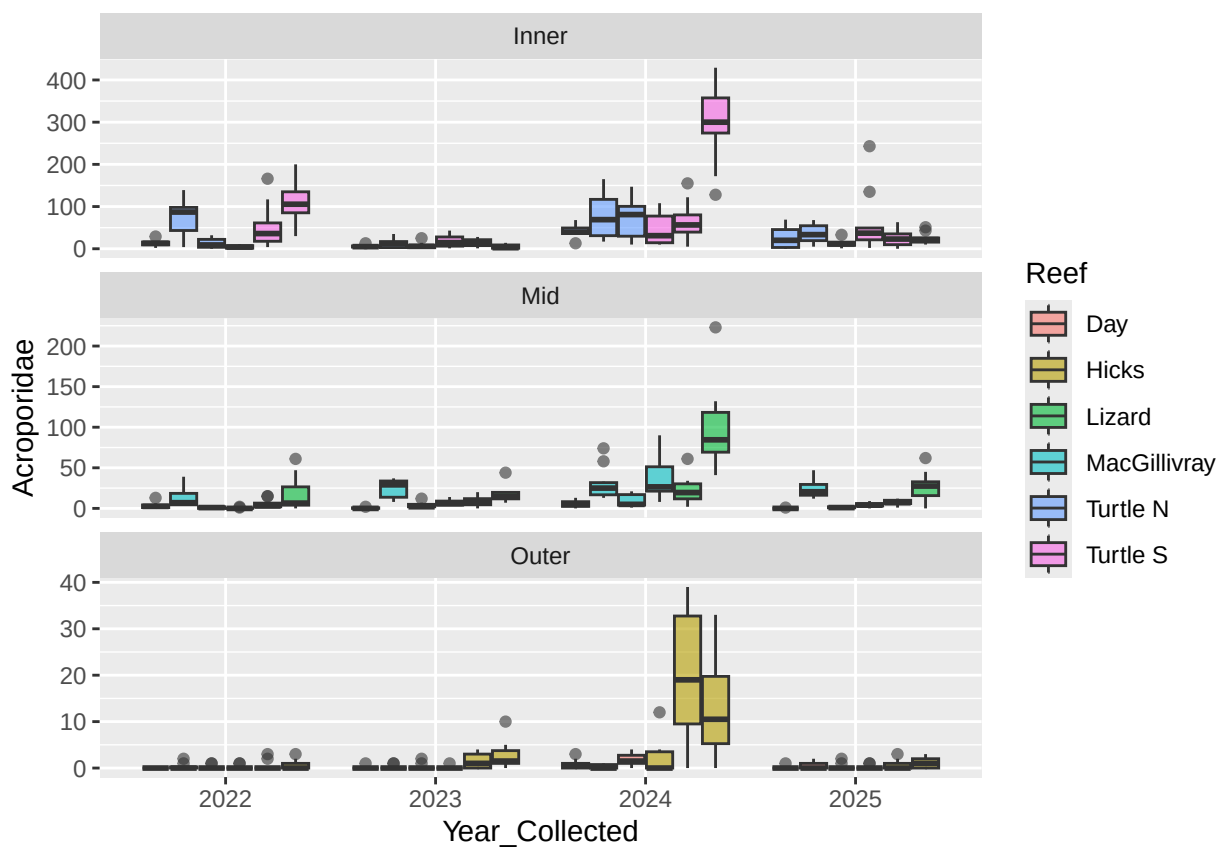
Negative Binomial Zero Inflated of Total recruits (per family)

```r
#Fixed
# Fit model with zero-inflated negative binomial distribution

zbin_Poritid_fixed <- glmmTMB(Poritid ~ Shelf_Position+ Year_Collected + (1 | Reef/Site) ,
                        ziformula = ~1,
                        family = "nbinom2",
                        data = Poritid)
zbin_Poritid_interaction <- glmmTMB(Poritid ~ Shelf_Position * Year_Collected + (1 | Reef/Site),
                        ziformula = ~1,
                        family = "nbinom2",
                        data = Poritid)

zbin_Pocilloporidae_fixed <- glmmTMB(Pocilloporidae ~ Shelf_Position+ Year_Collected + (1 | Reef/Site),
                               ziformula = ~1,
                        family = "nbinom2",
                        data = Pocilloporidae)
```

```
zbin_Pocilloporidae_interaction <- glmmTMB(Pocilloporidae ~ Shelf_Position*Year_Collected + (1 | Reef/Si
                                ziformula = ~1,
                          family = "nbinom2",
                          data = Pocilloporidae)

zbin_Acroporidae_fixed <- glmmTMB(Acroporidae ~ Shelf_Position + Year_Collected+ (1 | Reef/Site),
                              ziformula = ~1,
                          family = "nbinom2",
                          data = Acroporidae)
zbin_Acroporidae_interaction <- glmmTMB(Acroporidae ~ Shelf_Position*Year_Collected + (1 | Reef/Site),
                              ziformula = ~1,
                          family = "nbinom2",
                          data = Acroporidae)
```

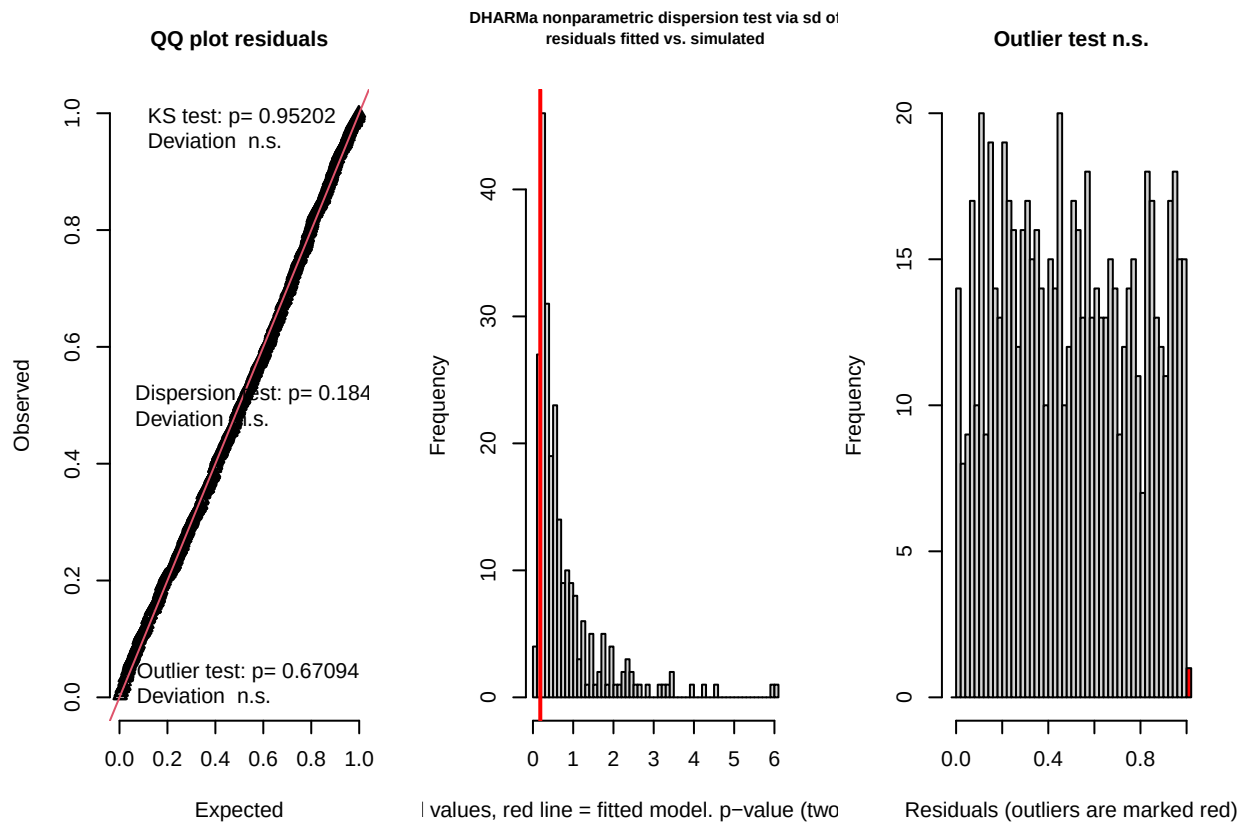
Acroporidae

```
Acroporidae |>
  ggplot(aes(x = Year_Collected, y = Acroporidae, fill = Reef)) +
  geom_boxplot(aes(group = interaction(Year_Collected, Site)),
               position = position_dodge(width = 0.8),
               alpha = 0.6) +
  facet_wrap(~Shelf_Position, ncol = 1, scale = "free_y")
```



Simulate residuals for models

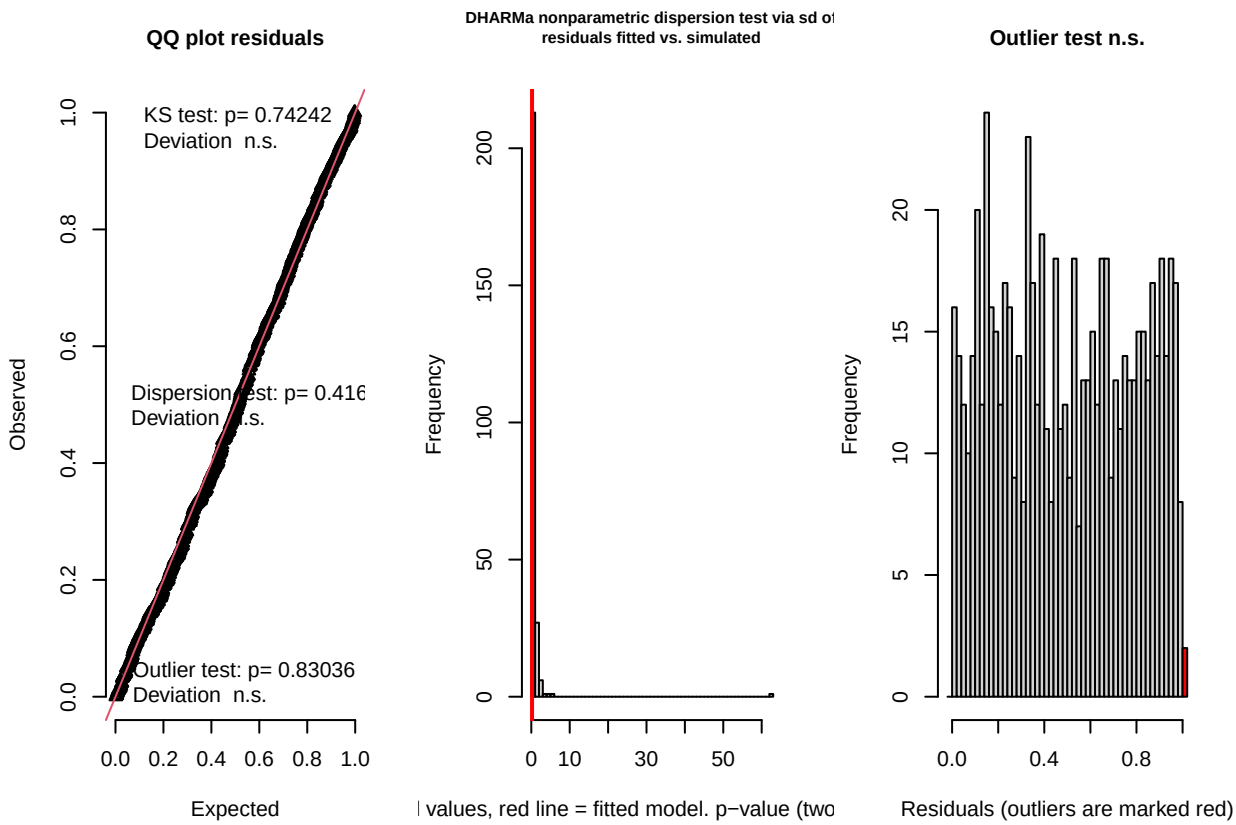
```r
sim2.1 <- simulateResiduals(nbin_Acroporidae_fixed)
testResiduals(sim2.1)
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.019444, p-value = 0.952
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.22598, p-value = 0.184
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
```

```
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 4, observations = 707, p-value = 0.6709
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.001543622 0.014422145
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.005657709
```

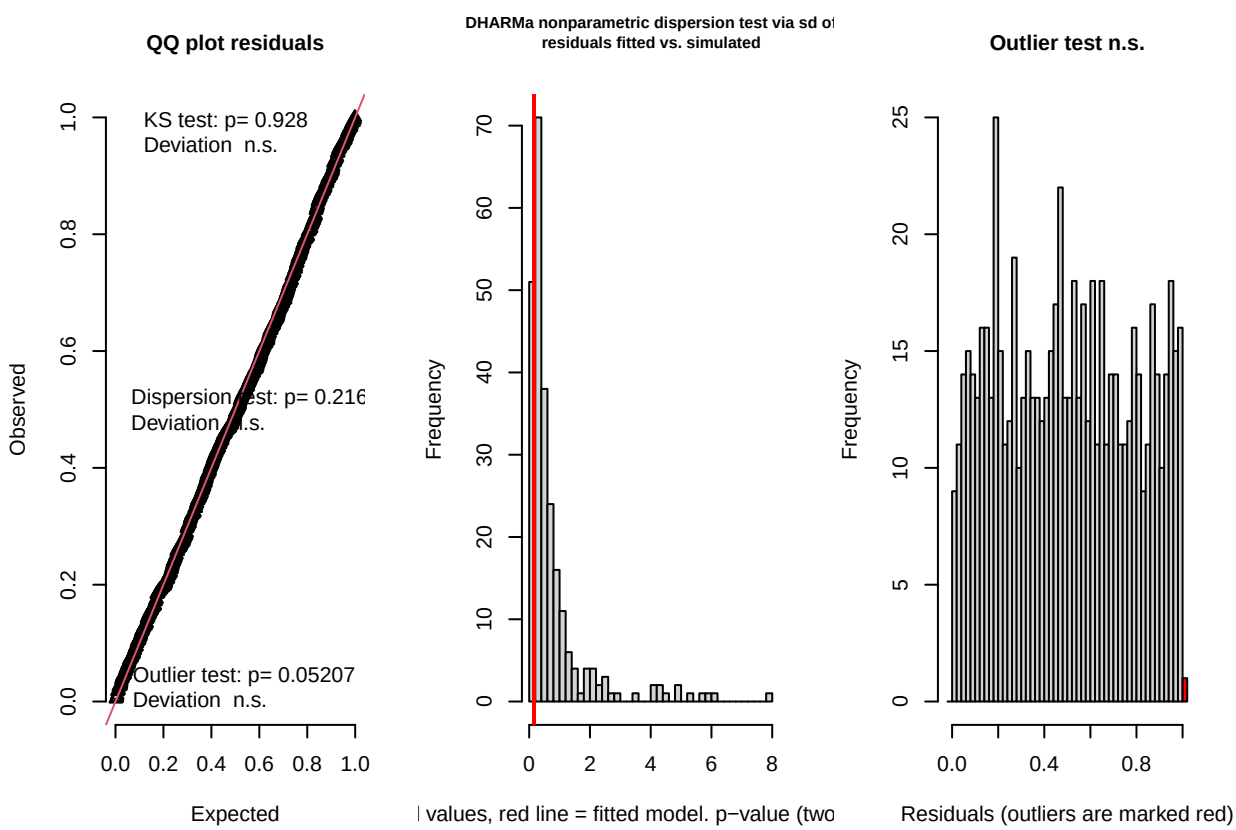
```
sim2.2 <- simulateResiduals(nbin_Acroporidae_interaction)
testResiduals(sim2.2)
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.025615, p-value = 0.7424
## alternative hypothesis: two-sided
##
##
## $dispersion
##
```

```
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.22847, p-value = 0.416
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 6, observations = 707, p-value = 0.8304
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.003120604 0.018379583
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.008486563
```

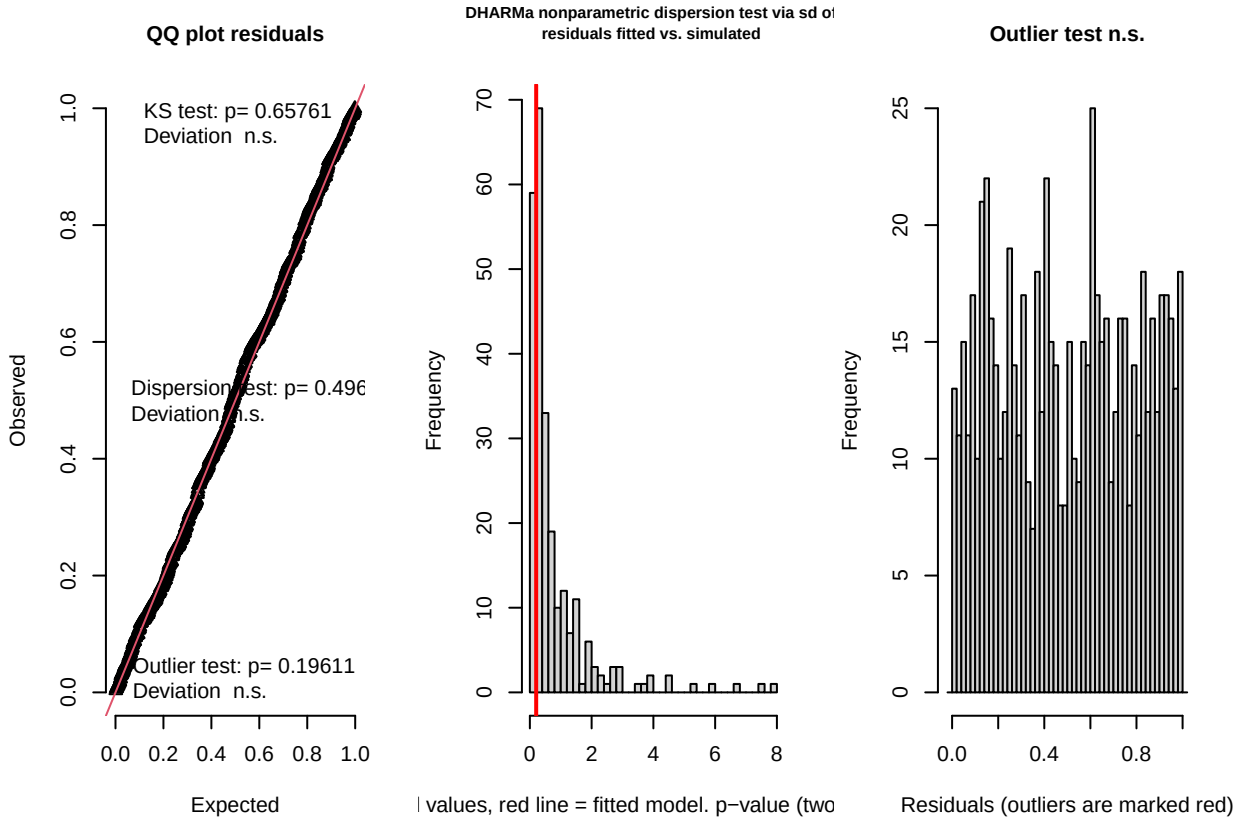
```
sim2.3 <- simulateResiduals(zbin_Acroporidae_fixed)
testResiduals(sim2.3)
```




```
## $uniformity
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.020487, p-value = 0.928
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.19199, p-value = 0.216
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 1, observations = 707, p-value = 0.05207
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  3.580955e-05 7.855252e-03
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                          0.001414427
```

```r
sim2.4 <- simulateResiduals(zbin_Acroporidae_interaction)
testResiduals(sim2.4)
```

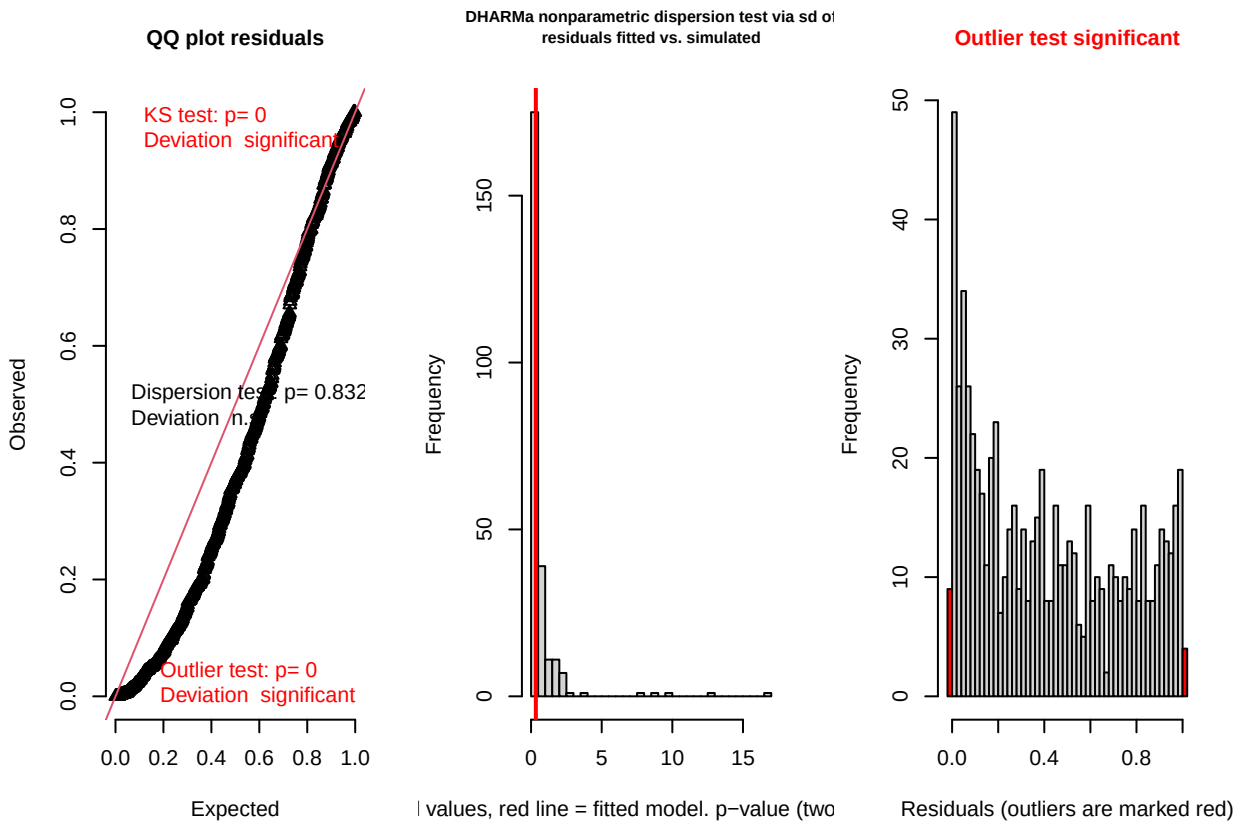



```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.027528, p-value = 0.6576
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.24769, p-value = 0.496
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 2, observations = 707, p-value = 0.1961
```

```
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.0003427712 0.0101811059
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.002828854
```

```
sim2.5 <- simulateResiduals(pois_Acroporidae_fixed) #not a good fit
testResiduals(sim2.5)
```

```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```

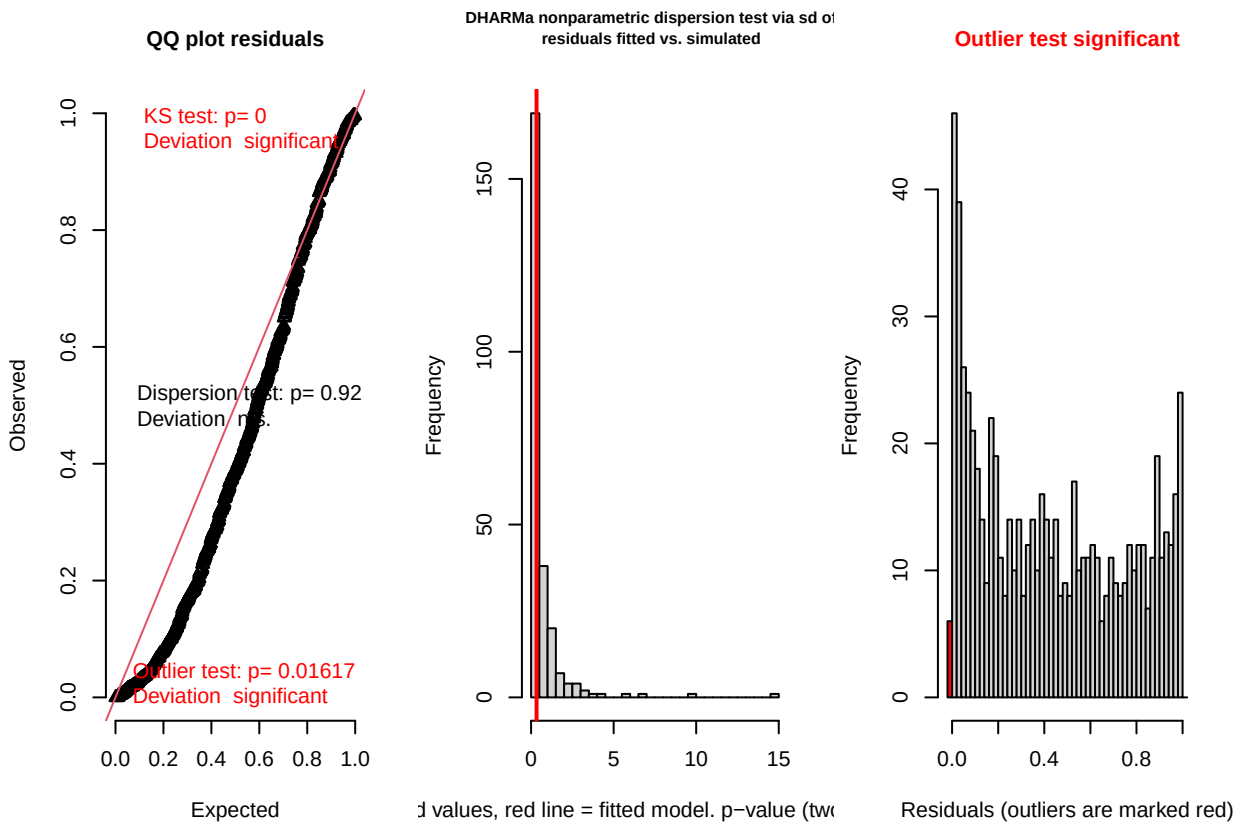



```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.1645, p-value < 2.2e-16
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
```

```
##  simulated
##
## data:  simulationOutput
## dispersion = 0.48303, p-value = 0.832
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 26, observations = 707, p-value =
## 2.778e-10
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.02416076 0.05342168
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.03677511
```

```r
sim2.6 <- simulateResiduals(pois_Acroporidae_interaction) #not a goodf fit
testResiduals(sim2.6)
```

```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.14601, p-value = 1.617e-13
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.48018, p-value = 0.92
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 12, observations = 707, p-value = 0.01617
```

```
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.008800231 0.029460782
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.01697313
```

Compare AIC of those models

```
AIC( nbin_Acroporidae_fixed, zbin_Acroporidae_fixed, nbin_Acroporidae_interaction, zbin_Acroporidae_int
```

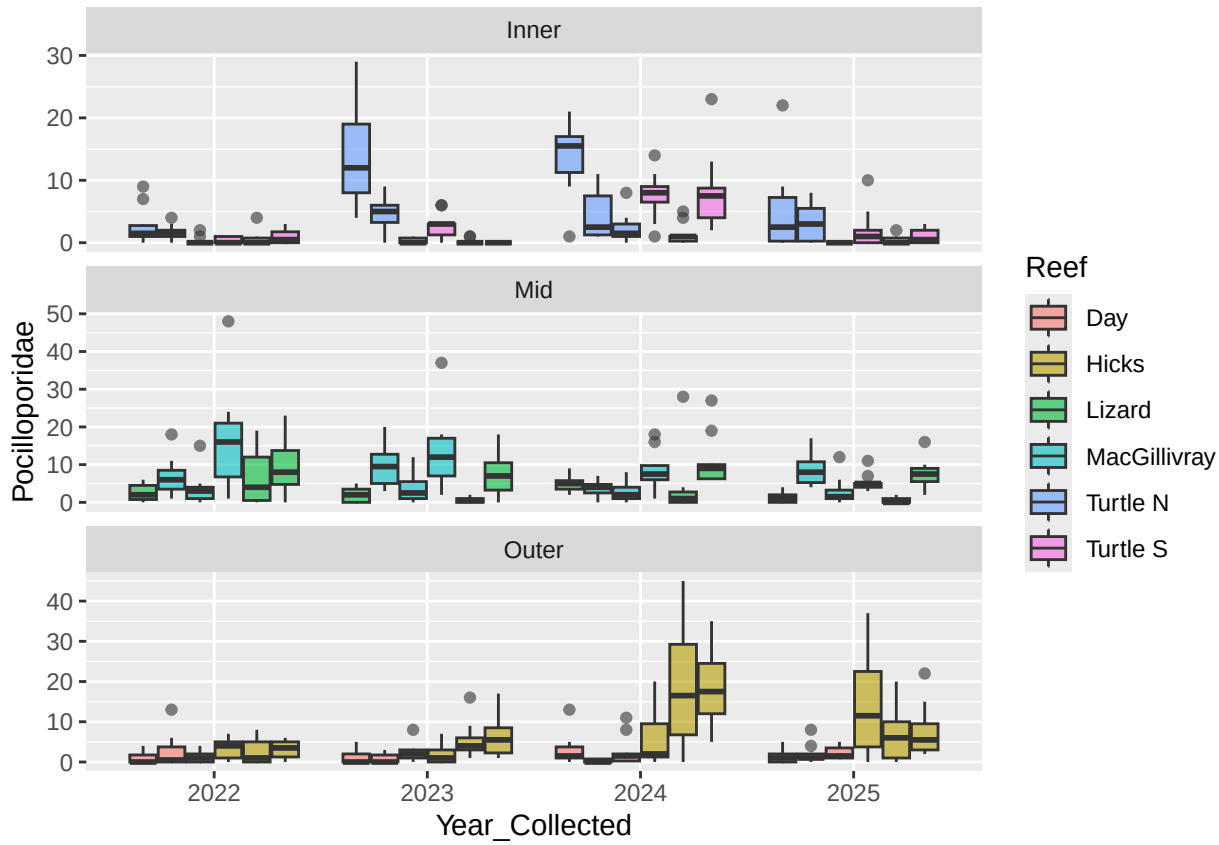
```
##                              df      AIC
## nbin_Acroporidae_fixed        9 4397.814
## zbin_Acroporidae_fixed       10 4399.814
## nbin_Acroporidae_interaction 15 4349.999
## zbin_Acroporidae_interaction 16 4350.570
```

```
##  Family: nbinom2  ( log )
## Formula:
## Acroporidae ~ Shelf_Position * Year_Collected + (1 | Reef/Site)
## Data: Acroporidae
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    4350.0    4418.4   -2160.0    4320.0       692
##
## Random effects:
##
## Conditional model:
##  Groups    Name        Variance  Std.Dev.
##  Site:Reef (Intercept) 7.456e-01 0.8634989
##  Reef      (Intercept) 4.386e-07 0.0006622
## Number of obs: 707, groups:  Site:Reef, 18; Reef, 6
##
## Dispersion parameter for nbinom2 family (): 1.16
##
## Conditional model:
##                                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)                              3.5805     0.3747   9.556  < 2e-16 ***
## Shelf_PositionMid                       -1.9108     0.5346  -3.574 0.000351 ***
## Shelf_PositionOuter                     -5.1083     0.5928  -8.618  < 2e-16 ***
## Year_Collected2023                      -1.2025     0.1847  -6.510 7.50e-11 ***
## Year_Collected2024                       0.8801     0.1751   5.026 5.00e-07 ***
## Year_Collected2025                      -0.1337     0.1839  -0.727 0.467213
## Shelf_PositionMid:Year_Collected2023     1.4553     0.2730   5.330 9.84e-08 ***
## Shelf_PositionOuter:Year_Collected2023   2.0237     0.3995   5.065 4.08e-07 ***
## Shelf_PositionMid:Year_Collected2024     0.6588     0.2624   2.511 0.012046 *
## Shelf_PositionOuter:Year_Collected2024   1.8009     0.3651   4.932 8.14e-07 ***
## Shelf_PositionMid:Year_Collected2025     0.1814     0.2707   0.670 0.502849
## Shelf_PositionOuter:Year_Collected2025   0.5941     0.4178   1.422 0.154979
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Acroporidae abundance significantly different in the outer reefs than inshore reefs.

Pocillopora

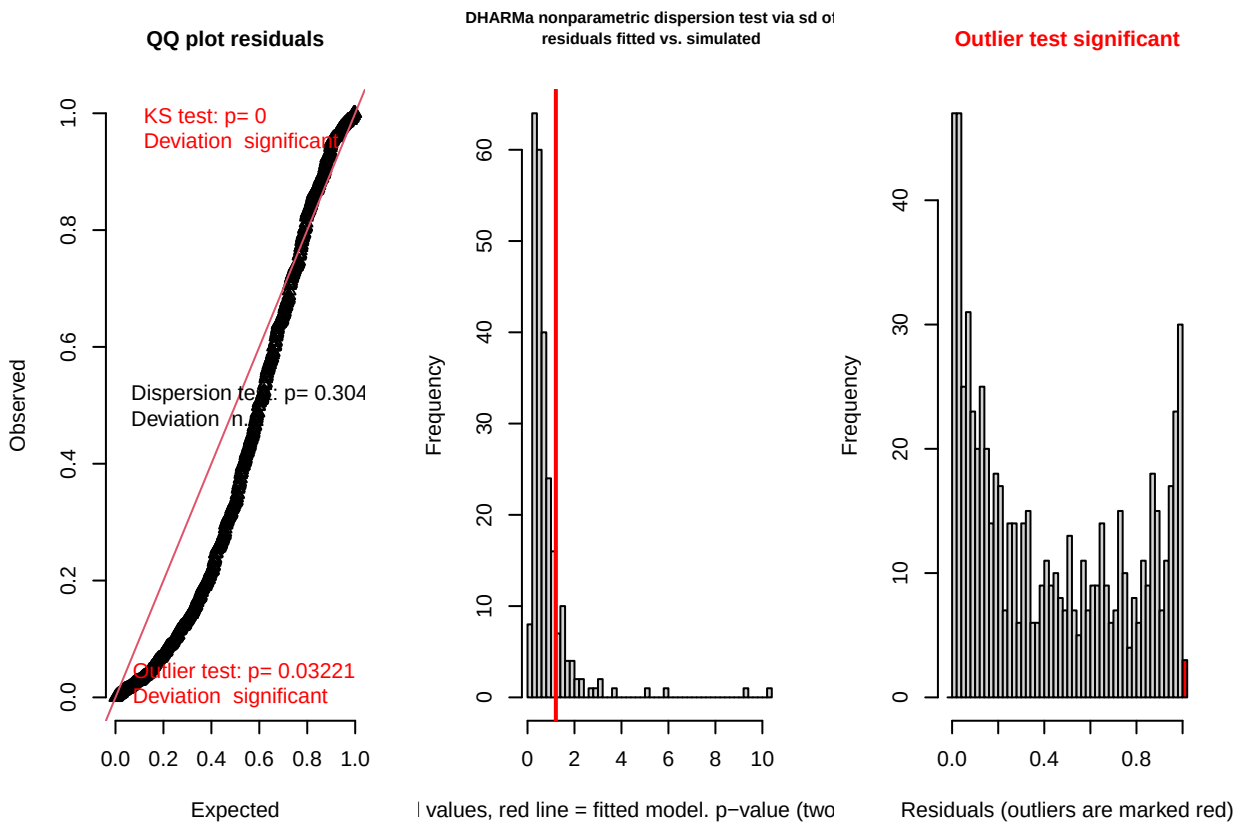
```
Pocilloporidae |>
  ggplot(aes(x = Year_Collected, y = Pocilloporidae, fill = Reef)) +
  geom_boxplot(aes(group = interaction(Year_Collected, Site)),
               position = position_dodge(width = 0.8),
               alpha = 0.6) +
  facet_wrap(~Shelf_Position, ncol = 1, scale = "free_y")
```



Test Residuals of models for Pocillopora

```
sim3.1 <- simulateResiduals(pois_Pocilloporidae_fixed)
testResiduals(sim3.1)
```

```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```

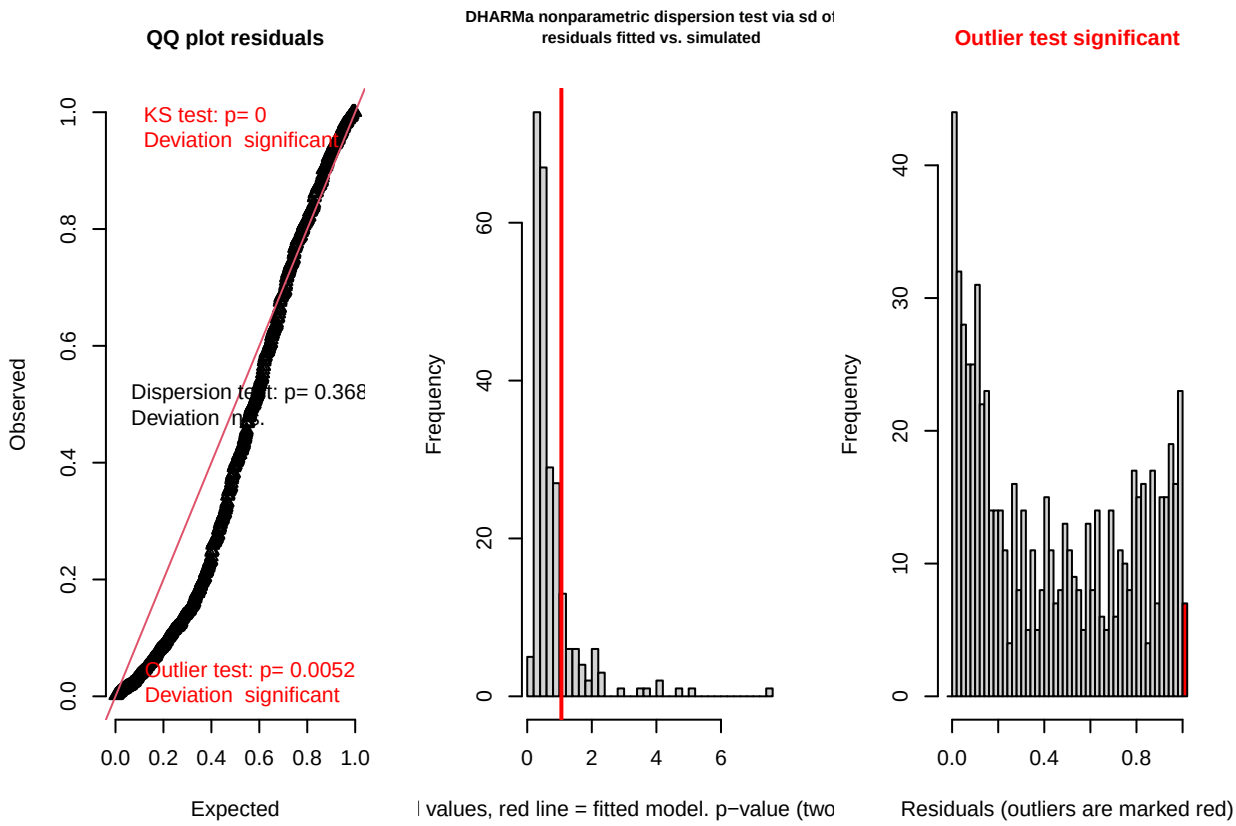



```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.18788, p-value < 2.2e-16
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 1.438, p-value = 0.304
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 11, observations = 707, p-value = 0.03221
```

```
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.007791767 0.027667962
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.0155587
```

```
sim3.2 <- simulateResiduals(pois_Pocilloporidae_interaction)
testResiduals(sim3.2)
```

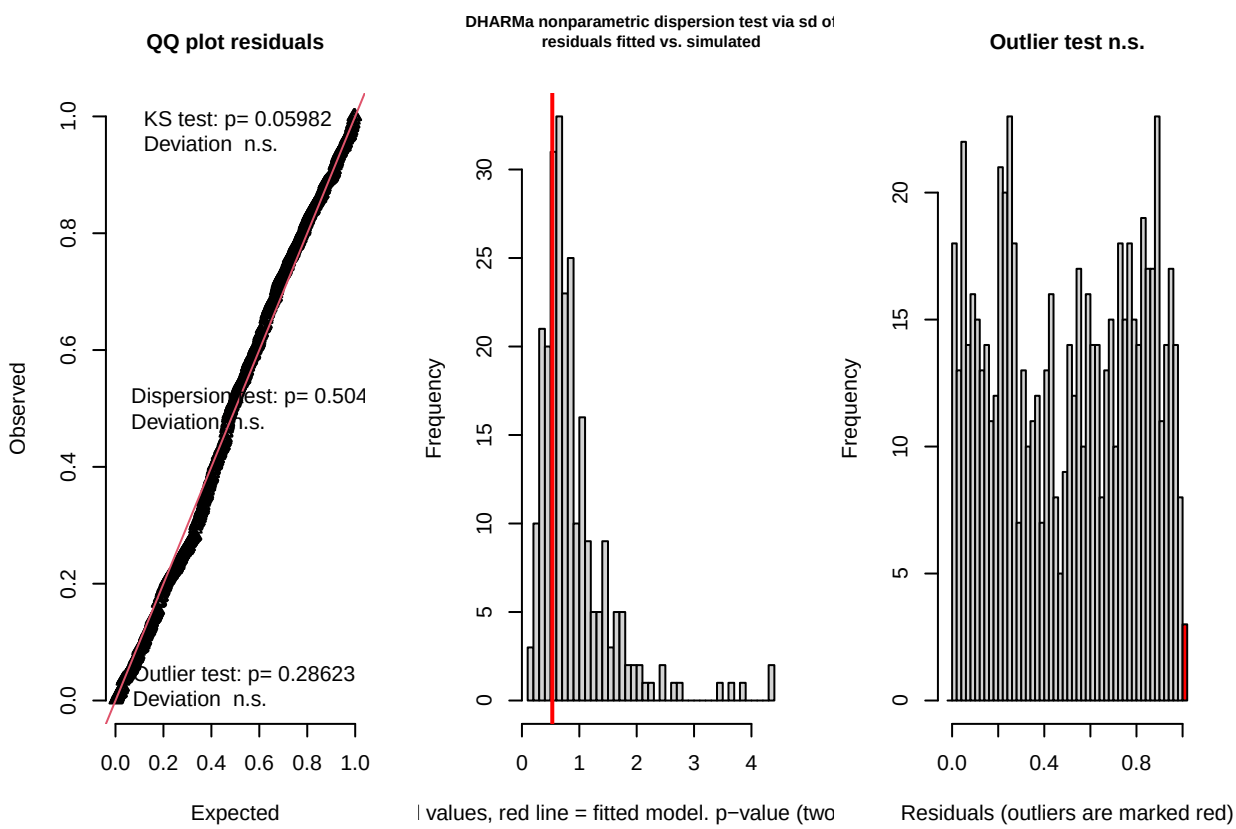
```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.16927, p-value < 2.2e-16
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
```

```
##  simulated
##
## data:  simulationOutput
## dispersion = 1.3348, p-value = 0.368
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 13, observations = 707, p-value = 0.005203
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.009826014 0.031238033
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.01838755
```

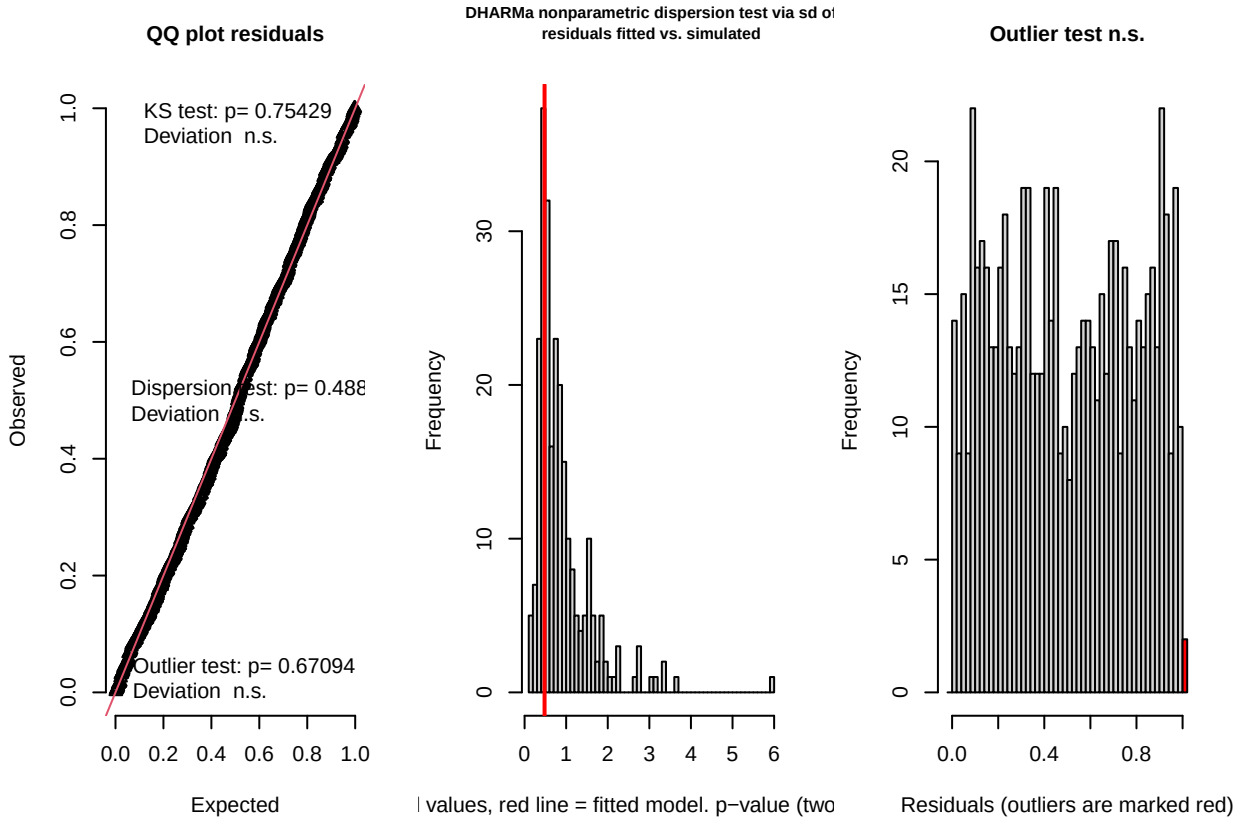
```
sim3.3 <- simulateResiduals(nbin_Pocilloporidae_interaction)
testResiduals(sim3.3)
```




```
## $uniformity
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.049819, p-value = 0.05982
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.58436, p-value = 0.504
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 8, observations = 707, p-value = 0.2862
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.004897502 0.022173604
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                              0.01131542
```

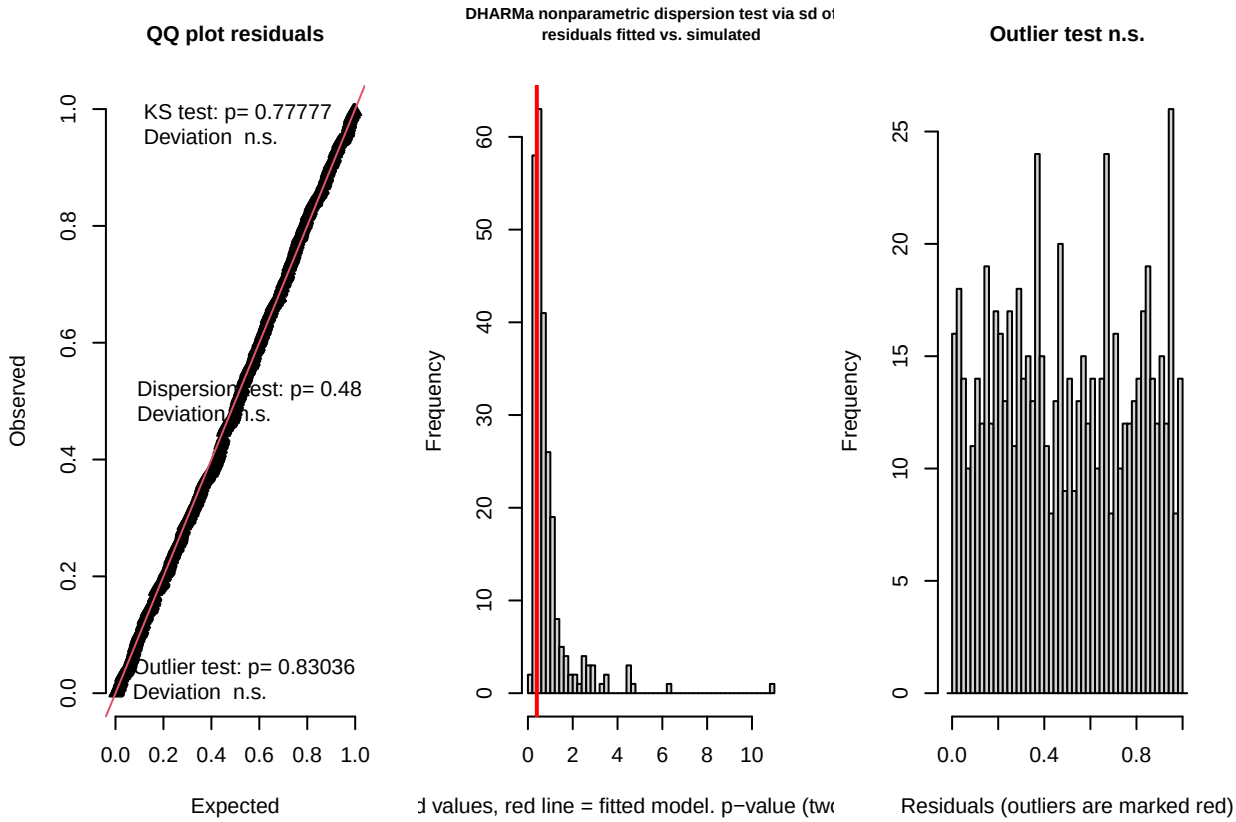
```r
sim3.4 <- simulateResiduals(nbin_Pocilloporidae_fixed)
testResiduals(sim3.4)
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.025341, p-value = 0.7543
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.52791, p-value = 0.488
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 4, observations = 707, p-value = 0.6709
```

```
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.001543622 0.014422145
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.005657709
```

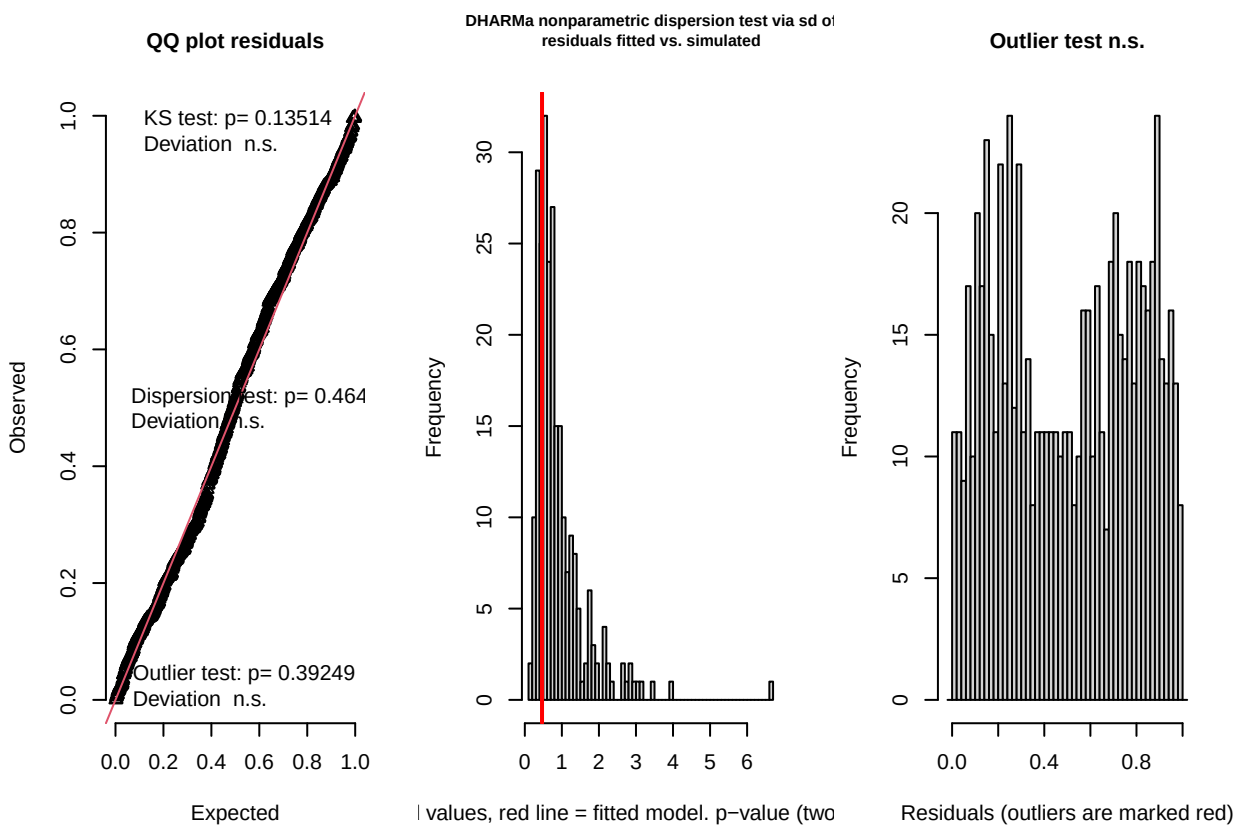
```
sim3.5 <- simulateResiduals(zbin_Pocilloporidae_fixed)
testResiduals(sim3.5)
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.024788, p-value = 0.7778
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
```

```
## dispersion = 0.4318, p-value = 0.48
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 6, observations = 707, p-value = 0.8304
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.003120604 0.018379583
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.008486563
```

```
sim3.6 <- simulateResiduals(zbin_Pocilloporidae_interaction)
testResiduals(sim3.6)
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
```

```
## data:  simulationOutput$scaledResiduals
## D = 0.043651, p-value = 0.1351
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.50602, p-value = 0.464
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 3, observations = 707, p-value = 0.3925
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.000875923 0.012350181
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                              0.004243281
```

```
AIC(pois_Pocilloporidae_fixed, nbin_Pocilloporidae_fixed, zbin_Pocilloporidae_fixed, nbin_Pocilloporidae
```

```
##                                df      AIC
## pois_Pocilloporidae_fixed       8 4950.318
## nbin_Pocilloporidae_fixed       9 3447.657
## zbin_Pocilloporidae_fixed      10 3449.657
## nbin_Pocilloporidae_interaction 15 3393.184
```

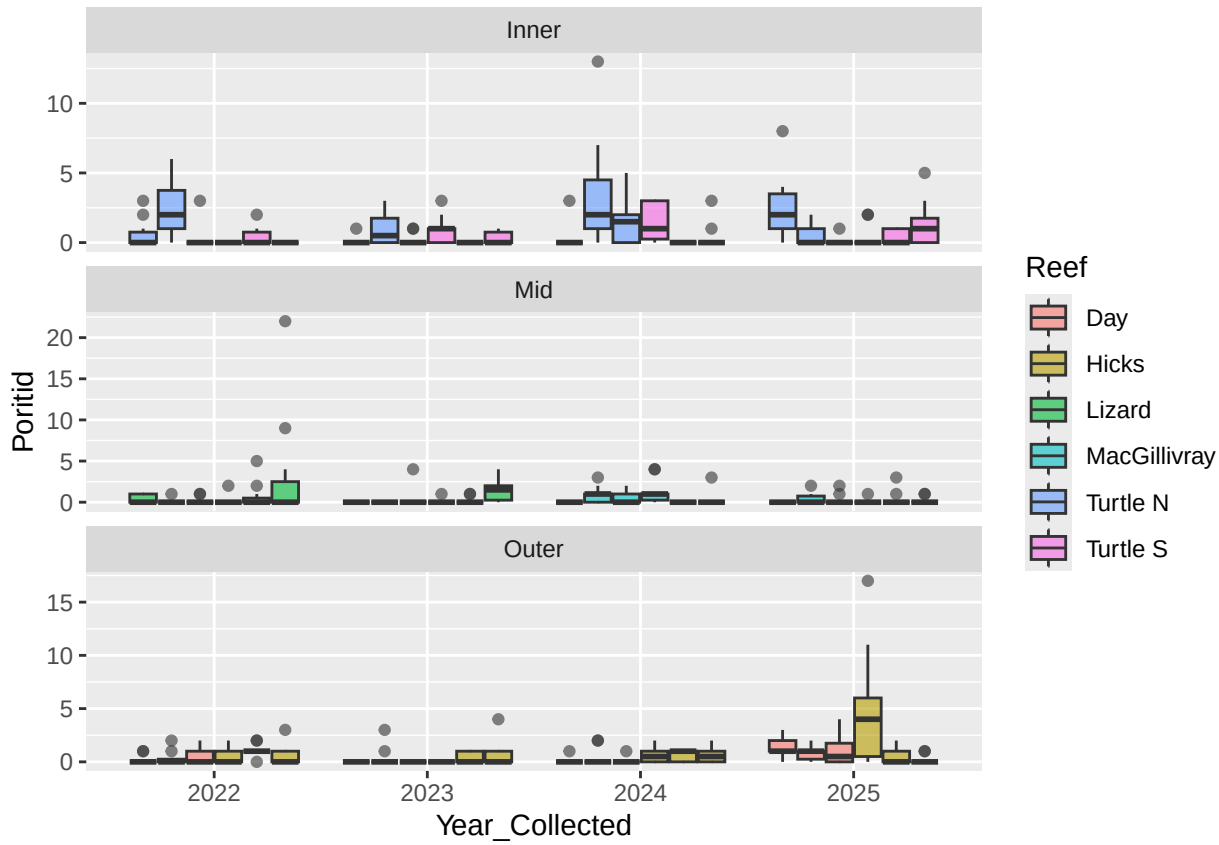
```
##  Family: nbinom2  ( log )
## Formula:
## Pocilloporidae ~ Shelf_Position * Year_Collected + (1 | Reef/Site)
## Data: Pocilloporidae
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    3393.2    3461.6   -1681.6    3363.2       692
##
## Random effects:
##
## Conditional model:
##  Groups    Name        Variance Std.Dev.
##  Site:Reef (Intercept) 0.52804  0.7267
##  Reef      (Intercept) 0.03277  0.1810
## Number of obs: 707, groups:  Site:Reef, 18; Reef, 6
##
```

```
## Dispersion parameter for nbinom2 family (): 1.23
##
## Conditional model:
##                                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)                             -0.2025     0.3743  -0.541  0.58853
## Shelf_PositionMid                        2.1466     0.5118   4.194 2.74e-05 ***
## Shelf_PositionOuter                      0.9895     0.5195   1.905  0.05679 .
## Year_Collected2023                       0.8375     0.2378   3.521  0.00043 ***
## Year_Collected2024                       1.8472     0.2332   7.920 2.38e-15 ***
## Year_Collected2025                       0.4410     0.2462   1.791  0.07328 .
## Shelf_PositionMid:Year_Collected2023    -1.2281     0.3044  -4.034 5.47e-05 ***
## Shelf_PositionOuter:Year_Collected2023  -0.7053     0.3238  -2.178  0.02940 *
## Shelf_PositionMid:Year_Collected2024    -2.0724     0.2988  -6.935 4.07e-12 ***
## Shelf_PositionOuter:Year_Collected2024  -0.9023     0.3130  -2.883  0.00394 **
## Shelf_PositionMid:Year_Collected2025    -1.1131     0.3127  -3.560  0.00037 ***
## Shelf_PositionOuter:Year_Collected2025   0.2869     0.3237   0.886  0.37542
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Significant difference in Pocilloporidae abundance in mid shore reefs compared to inshore reefs

Poritid

```r
Poritid |>
  ggplot(aes(x = Year_Collected, y = Poritid, fill = Reef)) +
  geom_boxplot(aes(group = interaction(Year_Collected, Site)),
               position = position_dodge(width = 0.8),
               alpha = 0.6) +
  facet_wrap(~Shelf_Position, ncol = 1, scale = "free_y")
```



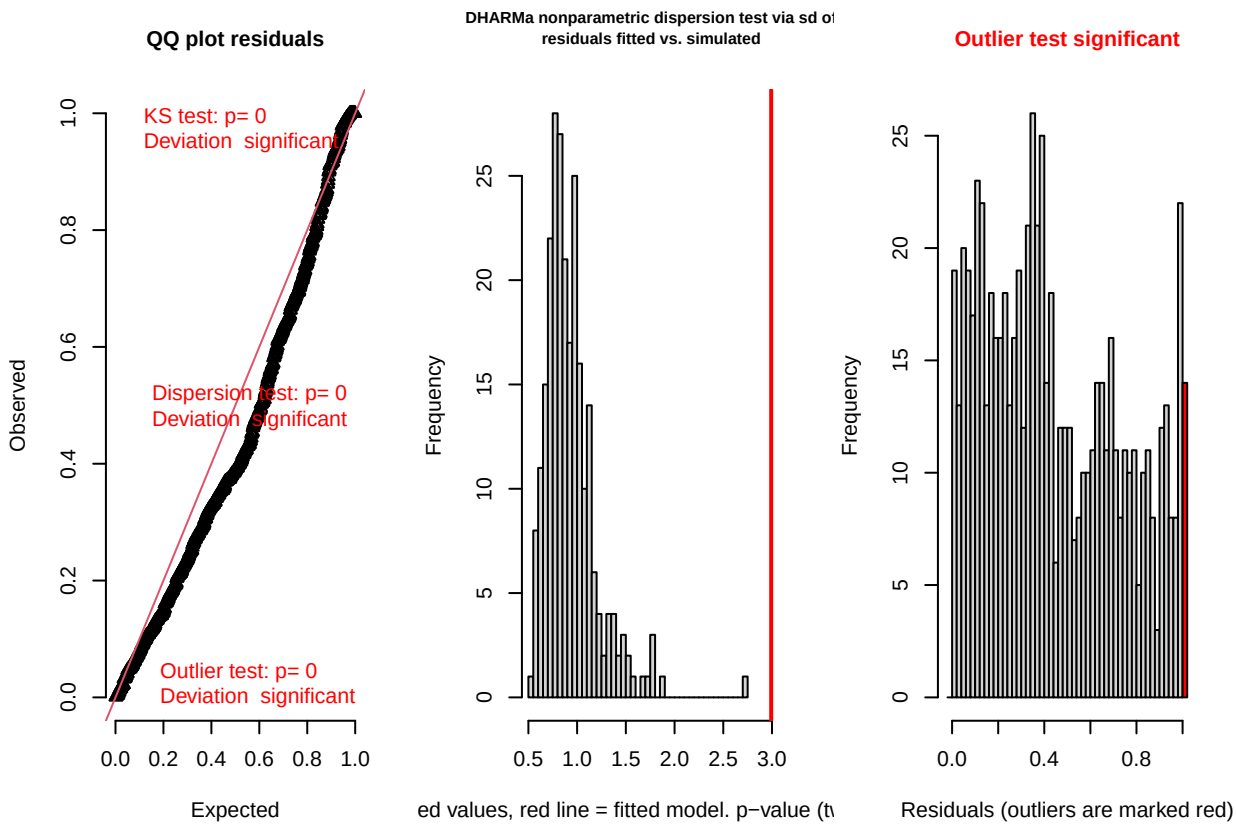

```
Poritid |>
  filter(Poritid>60)
```

```
## # A tibble: 0 x 10
## # i 10 variables: Year_Collected <fct>, Shelf_Position <fct>, Reef <chr>,
## #   Site <chr>, Tile <chr>, Poritid <dbl>, Top_Por <dbl>, Under_Por <dbl>,
## #   Side_Por <dbl>, total <dbl>
```

Test residuals of models

```
sim4.1 <- simulateResiduals(pois_Poritid_fixed)
testResiduals(sim4.1) # bad model
```

```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```

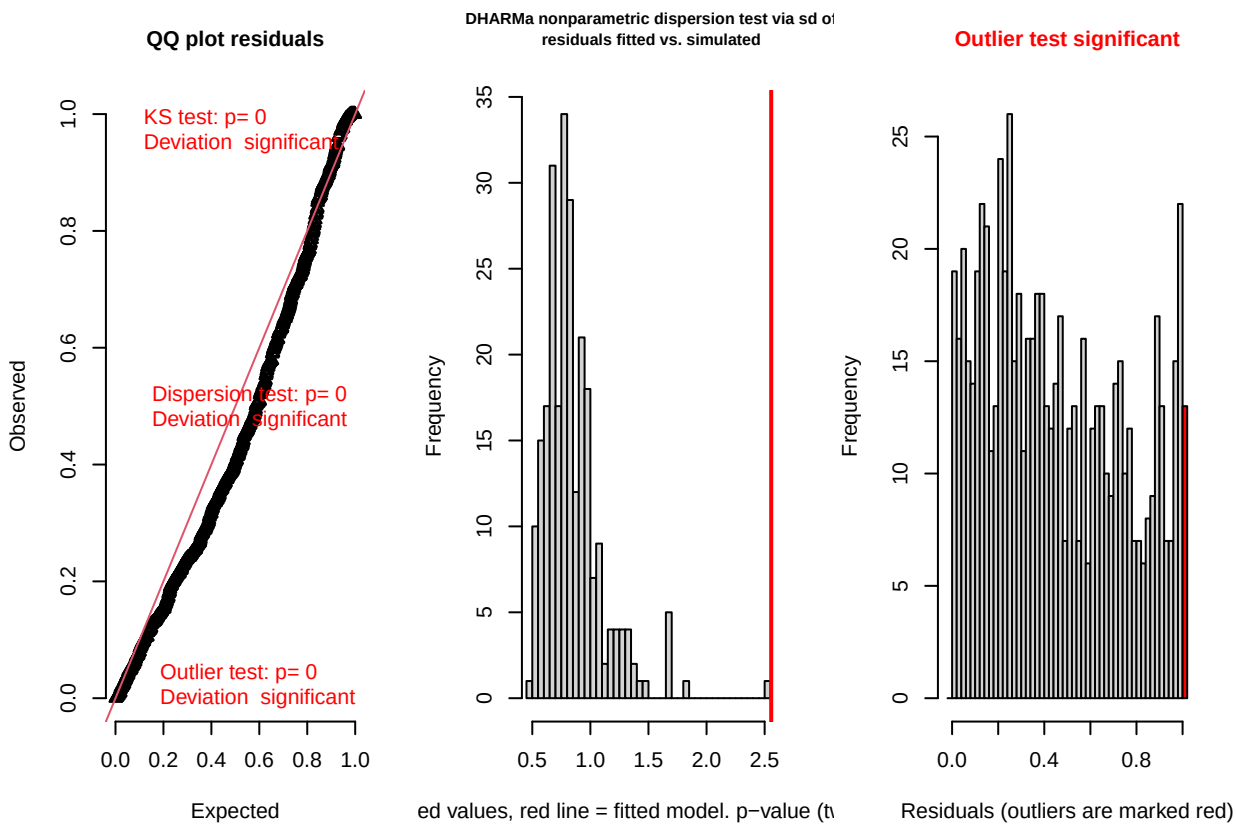



```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.12642, p-value = 3.071e-10
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 3.1976, p-value < 2.2e-16
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 23, observations = 707, p-value =
```

```
## 2.711e-08
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.02073192 0.04841541
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.03253182
```

```
sim4.2 <- simulateResiduals(pois_Poritid_interaction)
testResiduals(sim4.2) # bad model
```

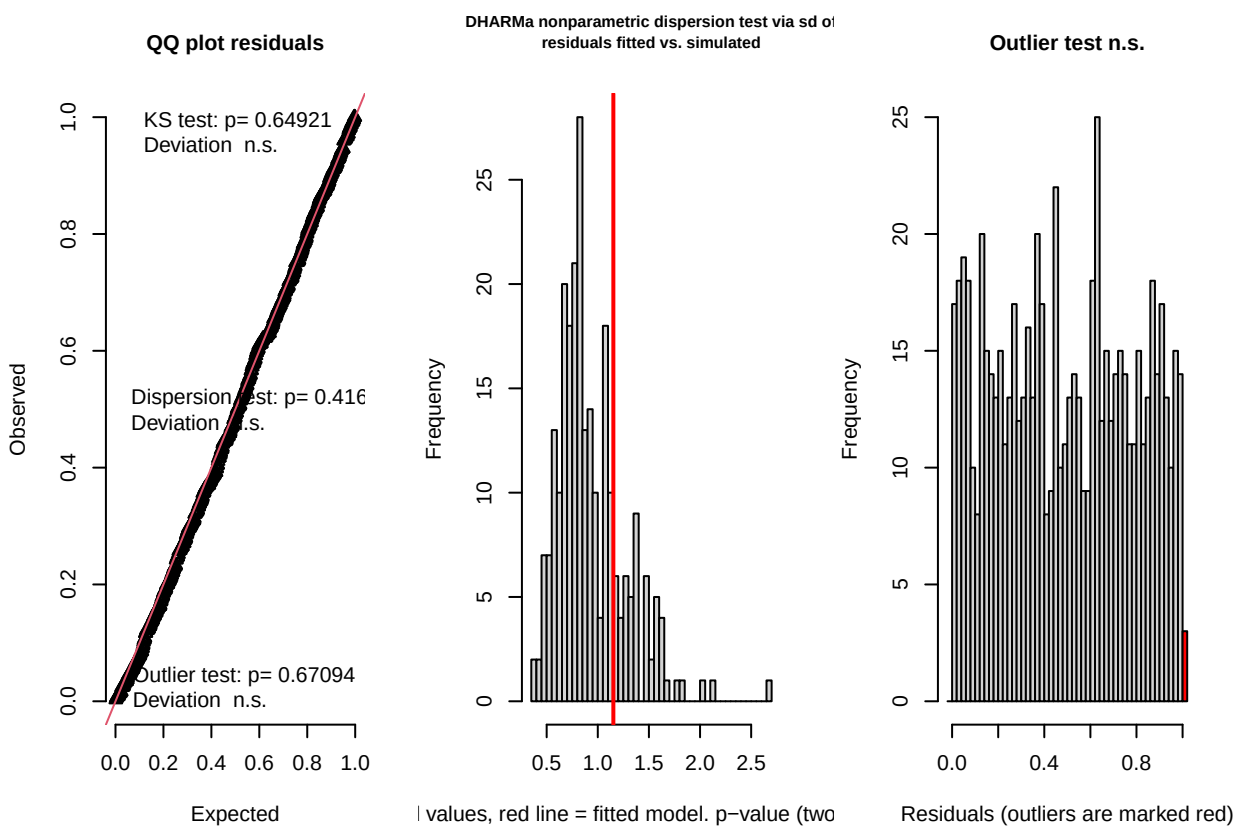
```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.099512, p-value = 1.659e-06
## alternative hypothesis: two-sided
##
##
## $dispersion
##
```

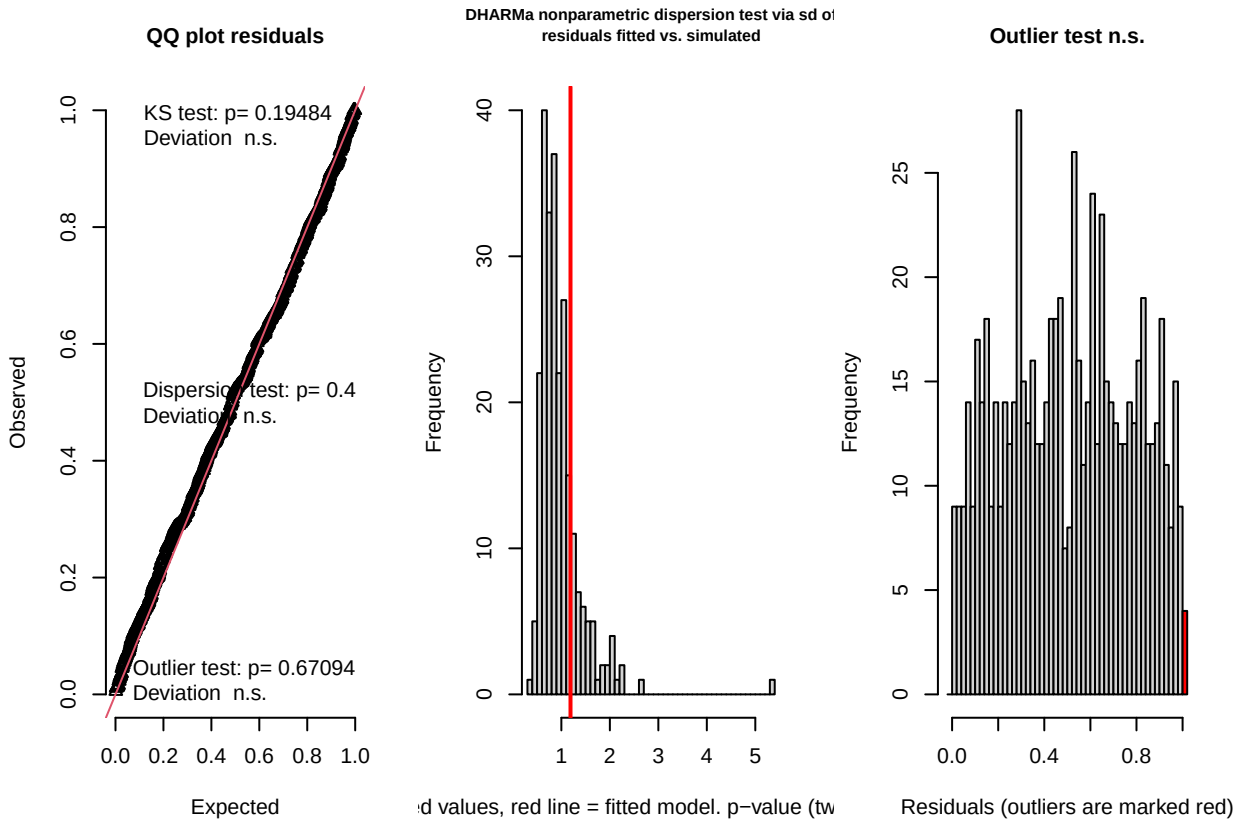
```
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 2.9909, p-value < 2.2e-16
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 25, observations = 707, p-value =
## 1.332e-09
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.02301181 0.05175862
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.03536068
```

```
sim4.3 <- simulateResiduals(nbin_Poritid_interaction)
testResiduals(sim4.3)
```



```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.027715, p-value = 0.6492
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 1.2308, p-value = 0.416
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 4, observations = 707, p-value = 0.6709
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.001543622 0.014422145
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                          0.005657709
```

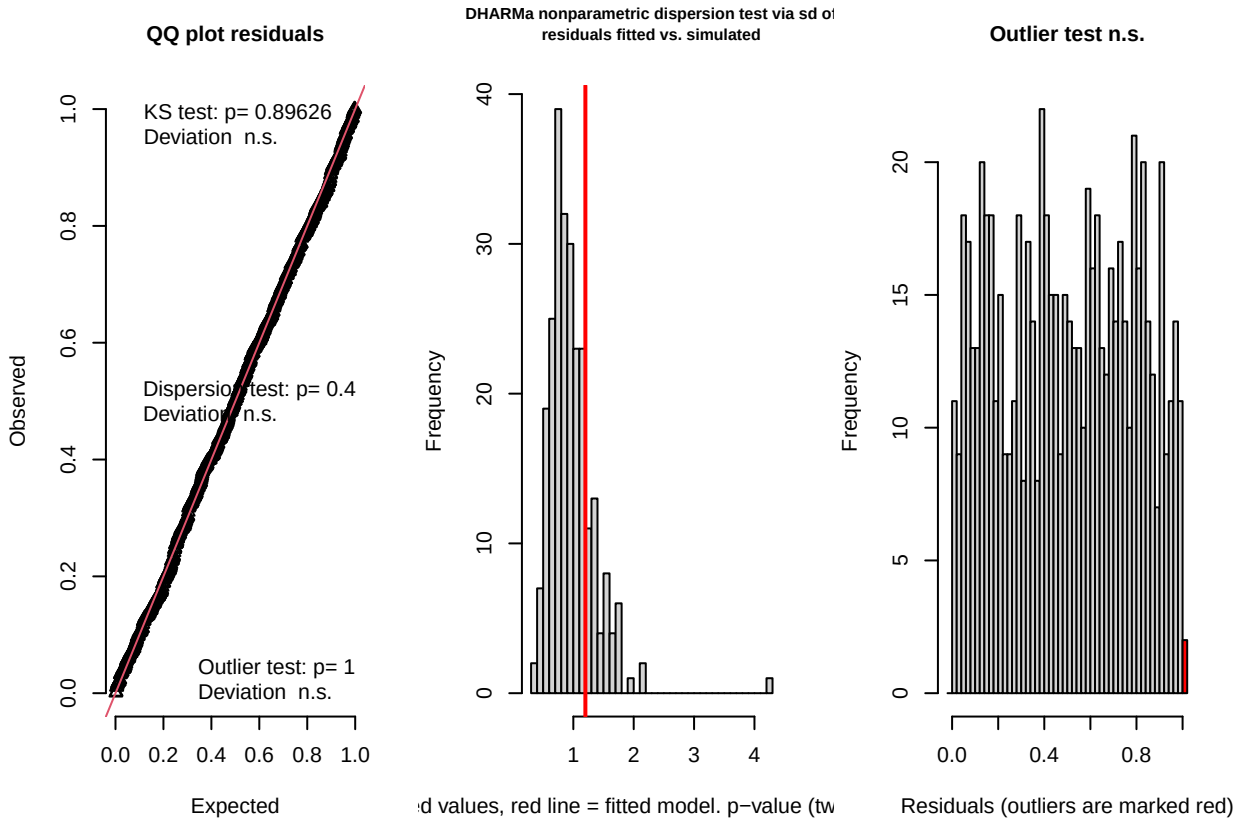
```r
sim4.4 <- simulateResiduals(nbin_Poritid_fixed)
testResiduals(sim4.4)
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.040574, p-value = 0.1948
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 1.2207, p-value = 0.4
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 4, observations = 707, p-value = 0.6709
```

```
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.001543622 0.014422145
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.005657709
```

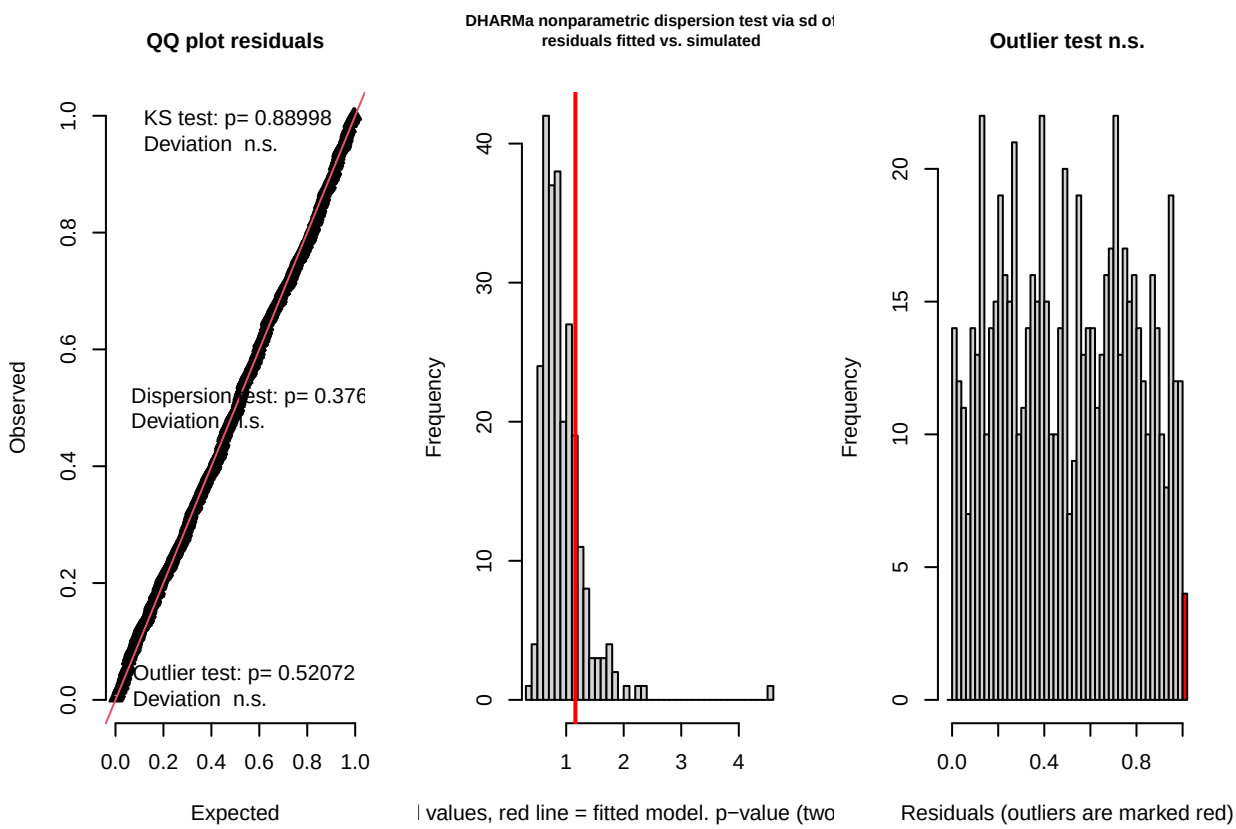
```
sim4.5 <- simulateResiduals(zbin_Poritid_fixed)
testResiduals(sim4.5)
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.021602, p-value = 0.8963
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
```

```
## dispersion = 1.2323, p-value = 0.4
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 5, observations = 707, p-value = 1
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.002300174 0.016426295
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.007072136
```

```
sim4.6 <- simulateResiduals(zbin_Poritid_interaction)
testResiduals(sim4.6)
```




```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
```

```
## data:  simulationOutput$scaledResiduals
## D = 0.021801, p-value = 0.89
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 1.2426, p-value = 0.376
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 7, observations = 707, p-value = 0.5207
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.003989716 0.020292869
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.00990099

##                          df      AIC
## pois_Poritid_fixed        8 1764.922
## nbin_Poritid_fixed        9 1472.715
## zbin_Poritid_fixed       10 1474.715
## nbin_Poritid_interaction 15 1456.760
## zbin_Poritid_interaction 16 1458.760
## pois_Poritid_interaction 14 1692.555

##  Family: nbinom2  ( log )
## Formula:          Poritid ~ Shelf_Position * Year_Collected + (1 | Reef/Site)
## Data: Poritid
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    1456.8    1525.2    -713.4    1426.8       692
##
## Random effects:
##
## Conditional model:
##  Groups    Name        Variance  Std.Dev.
##  Site:Reef (Intercept) 2.721e-01 0.5216144
##  Reef      (Intercept) 1.923e-08 0.0001387
## Number of obs: 707, groups:  Site:Reef, 18; Reef, 6
##
## Dispersion parameter for nbinom2 family (): 0.575
##
```

```
## Conditional model:
##                                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)                             -0.7471     0.3340  -2.237 0.025302 *
## Shelf_PositionMid                        0.4145     0.4637   0.894 0.371338
## Shelf_PositionOuter                      0.1604     0.4723   0.340 0.734046
## Year_Collected2023                      -0.3281     0.3754  -0.874 0.382130
## Year_Collected2024                       0.7215     0.3331   2.166 0.030299 *
## Year_Collected2025                       0.5731     0.3497   1.639 0.101278
## Shelf_PositionMid:Year_Collected2023    -0.6021     0.5262  -1.144 0.252530
## Shelf_PositionOuter:Year_Collected2023  -0.5265     0.5607  -0.939 0.347693
## Shelf_PositionMid:Year_Collected2024    -1.1275     0.4944  -2.280 0.022585 *
## Shelf_PositionOuter:Year_Collected2024  -1.0945     0.4990  -2.193 0.028292 *
## Shelf_PositionMid:Year_Collected2025    -1.7911     0.5357  -3.343 0.000828 ***
## Shelf_PositionOuter:Year_Collected2025   0.3049     0.4872   0.626 0.531483
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Both nbin_interaction, and zbin_interaction for Portidae have significant dispersion. Don't see much of a difference in Poritid across shelf position. Although slightly less in mid shore reefs (p<0.1)

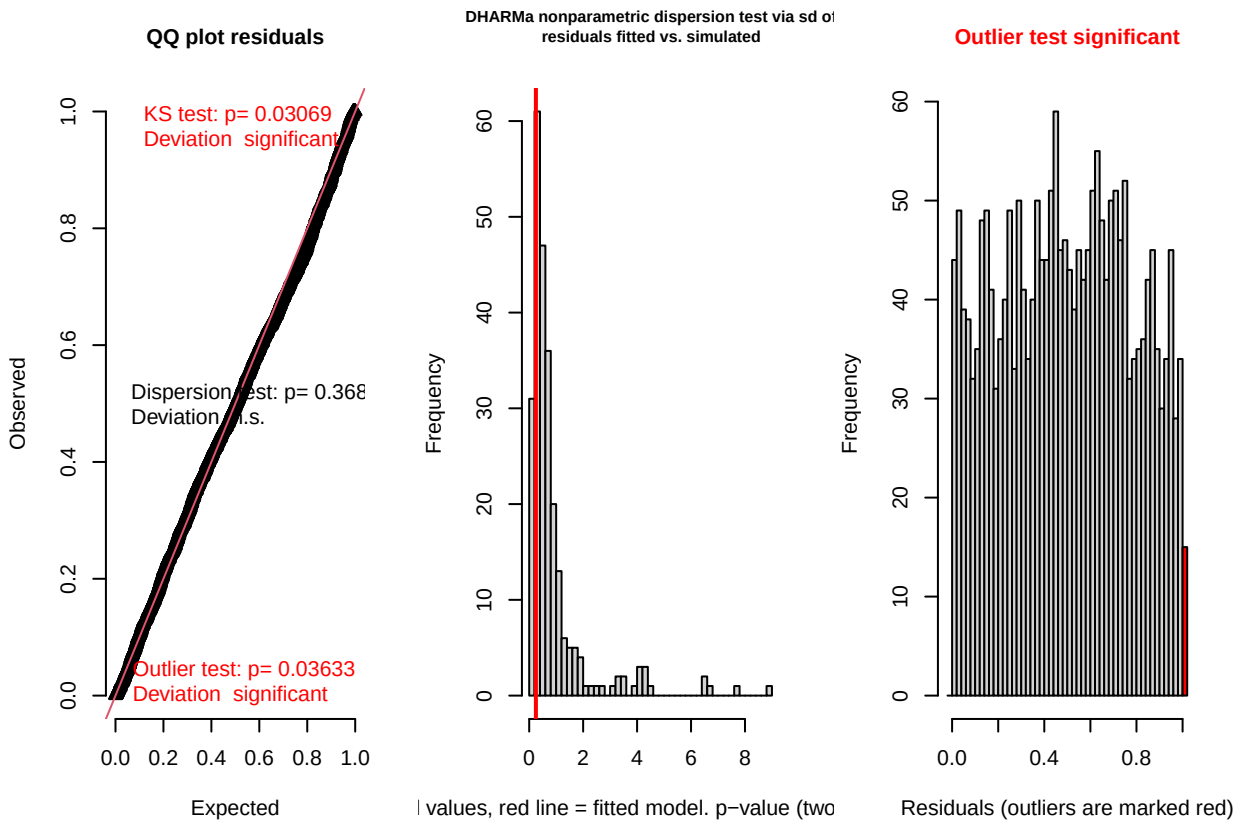
```r
nb_Orientation_Acroporidae <- glmmTMB(Recruits ~ 0+Shelf_Position* Tile_Surface + (1 | Reef/Site),
        family = "nbinom2", data = Acroporidae_orientation)

pois_Orientation_Acro <- glmmTMB(Recruits ~ 0+Shelf_Position* Tile_Surface + (1 | Reef/Site),
        family = "poisson", data = Acroporidae_orientation)

zi_orientation_Acro <- glmmTMB(Recruits ~ 0+Shelf_Position* Tile_Surface + (1 | Reef/Site),
                               ziformula = ~1,
        family = "nbinom2", data = Acroporidae_orientation)
```

AIC to determine model of best fit

```
##                             df      AIC
## nb_Orientation_Acroporidae 12  8369.42
## pois_Orientation_Acro      11 28326.16
## zi_orientation_Acro        13  8371.42
```

```
##  Family: nbinom2  ( log )
## Formula:
## Recruits ~ 0 + Shelf_Position * Tile_Surface + (1 | Reef/Site)
## Data: Acroporidae_orientation
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    8369.4    8437.3   -4172.7    8345.4      2109
##
## Random effects:
##
## Conditional model:
##  Groups    Name        Variance  Std.Dev.
##  Site:Reef (Intercept) 8.512e-01 0.922631
##  Reef      (Intercept) 1.512e-06 0.001229
## Number of obs: 2121, groups:  Site:Reef, 18; Reef, 6
##
```

```
## Dispersion parameter for nbinom2 family (): 0.399
##
## Conditional model:
##                                      Estimate Std. Error z value Pr(>|z|)
## Shelf_PositionInner                    2.6822     0.3921   6.841 7.87e-12 ***
## Shelf_PositionMid                      2.2652     0.3925   5.772 7.84e-09 ***
## Shelf_PositionOuter                   -0.1293     0.4011  -0.322   0.7472
## Tile_SurfaceSide                       0.3781     0.1535   2.464   0.0137 *
## Tile_SurfaceTop                       -1.1603     0.1582  -7.336 2.20e-13 ***
## Shelf_PositionMid:Tile_SurfaceSide    -2.6465     0.2311 -11.454  < 2e-16 ***
## Shelf_PositionOuter:Tile_SurfaceSide  -2.0962     0.2771  -7.565 3.89e-14 ***
## Shelf_PositionMid:Tile_SurfaceTop     -2.0459     0.2475  -8.267  < 2e-16 ***
## Shelf_PositionOuter:Tile_SurfaceTop   -1.9950     0.3601  -5.540 3.02e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```

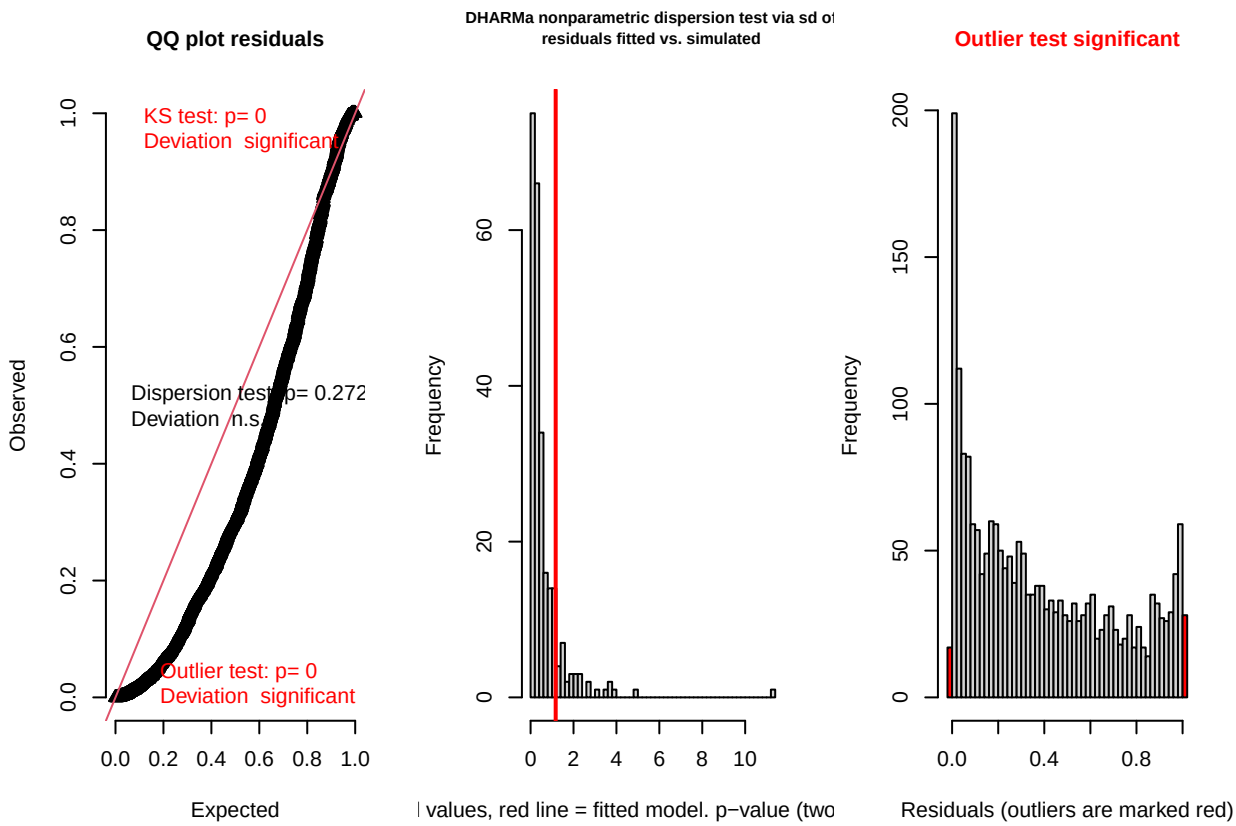



```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.03138, p-value = 0.03069
```

```
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.26928, p-value = 0.368
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 26, observations = 2121, p-value = 0.03633
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.008022773 0.017910082
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.01225837
```

```r
sim5.1 <- simulateResiduals(pois_Orientation_Acro)
testResiduals(sim5.1)
```

```
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
## DHARMa:testOutliers with type = binomial may have inflated Type I error rates for integer-valued dis
```



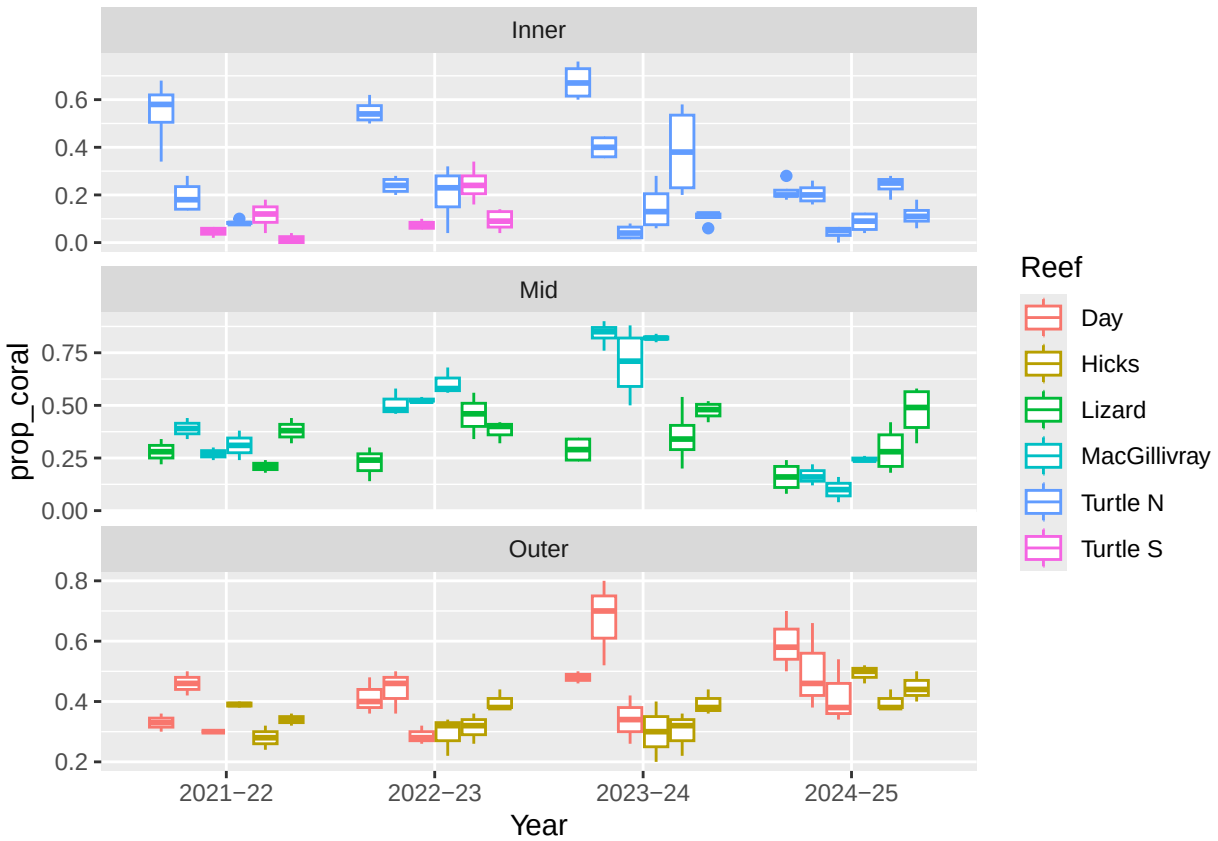

```
## $uniformity
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  simulationOutput$scaledResiduals
## D = 0.20116, p-value < 2.2e-16
## alternative hypothesis: two-sided
##
##
## $dispersion
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 1.8081, p-value = 0.272
## alternative hypothesis: two.sided
##
##
## $outliers
##
##  DHARMa outlier test based on exact binomial test with approximate
##  expectations
##
## data:  simulationOutput
## outliers at both margin(s) = 113, observations = 2121, p-value <
```

```
## 2.2e-16
## alternative hypothesis: true probability of success is not equal to 0.007968127
## 95 percent confidence interval:
##  0.04410627 0.06370490
## sample estimates:
## frequency of outliers (expected: 0.00796812749003984 )
##                                                0.05327676
```

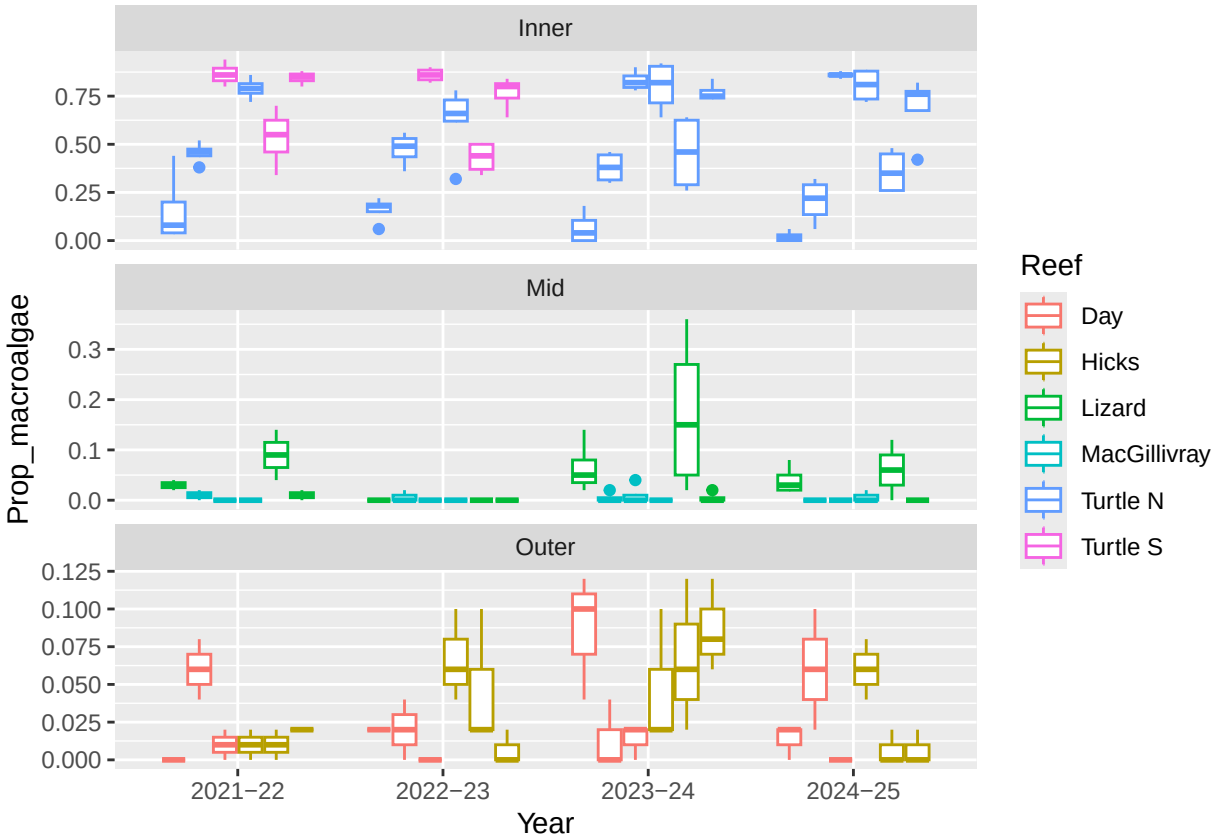
Brood Stock Analysis

Check whether benthic composition is behaving consistently between Sites in a reef

```
PIT_50m <- read_csv("PIT_50m.csv", col_types = cols(Shelf = col_character()))

PIT_50m <- PIT_50m |>
  mutate(
    Shelf = factor(Shelf),
    Reef = factor(Reef),
    Site = factor(Site)
         ) |>
  filter(Year %in% c("2021-22", "2022-23", "2023-24", "2024-25"))

PIT_50m <- PIT_50m |>
  mutate(
    Acroporidae = Acroporidae/50,
    Pocilloporidae = Pocilloporidae/50,
    Poritidae = Poritidae/50
  )



ggplot(PIT_50m, aes(x=Year, y=prop_coral, colour = Reef))+
  geom_boxplot(aes(group = interaction(Year, Site))) +
  facet_wrap(~Shelf, ncol = 1, scale = "free_y")
```

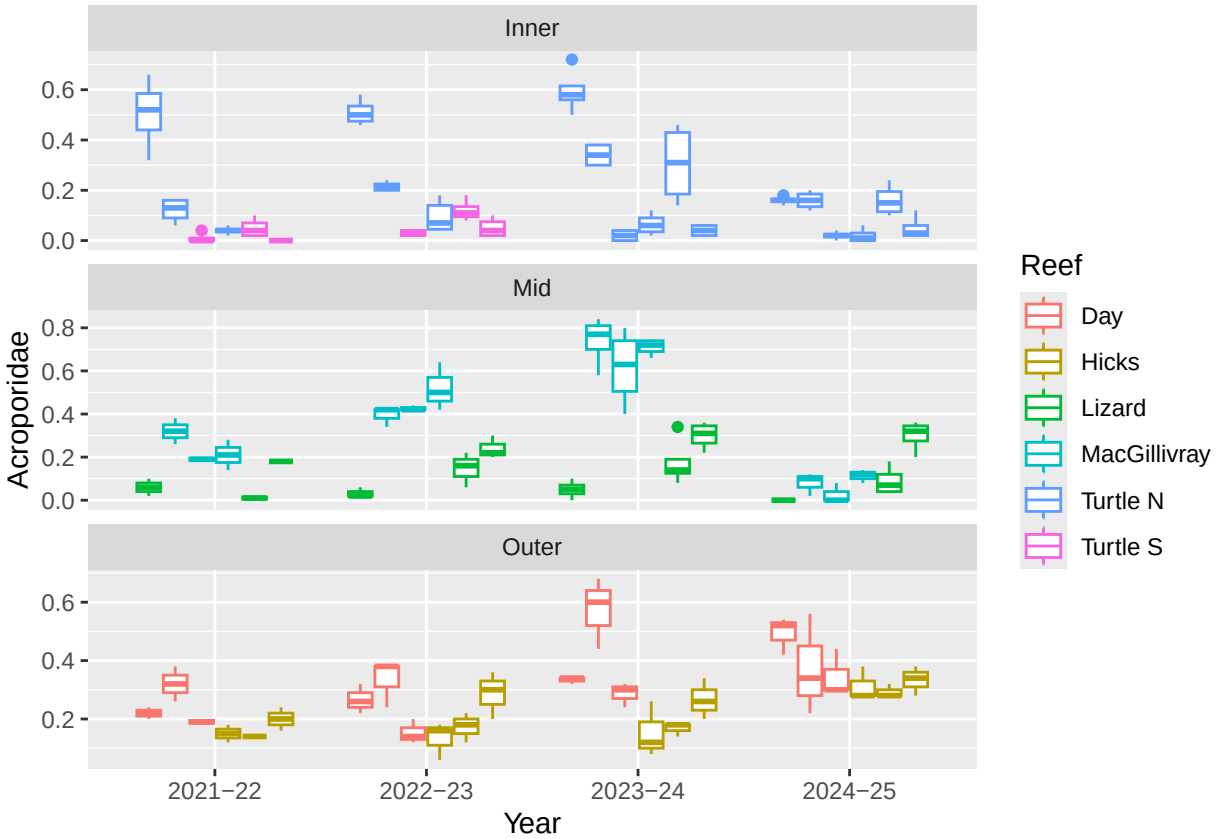



```
ggplot(PIT_50m, aes(x=Year, y=Prop_macroalgae, colour = Reef))+
  geom_boxplot(aes(group = interaction(Year, Site))) +
  facet_wrap(~Shelf, ncol = 1, scale = "free_y")
```



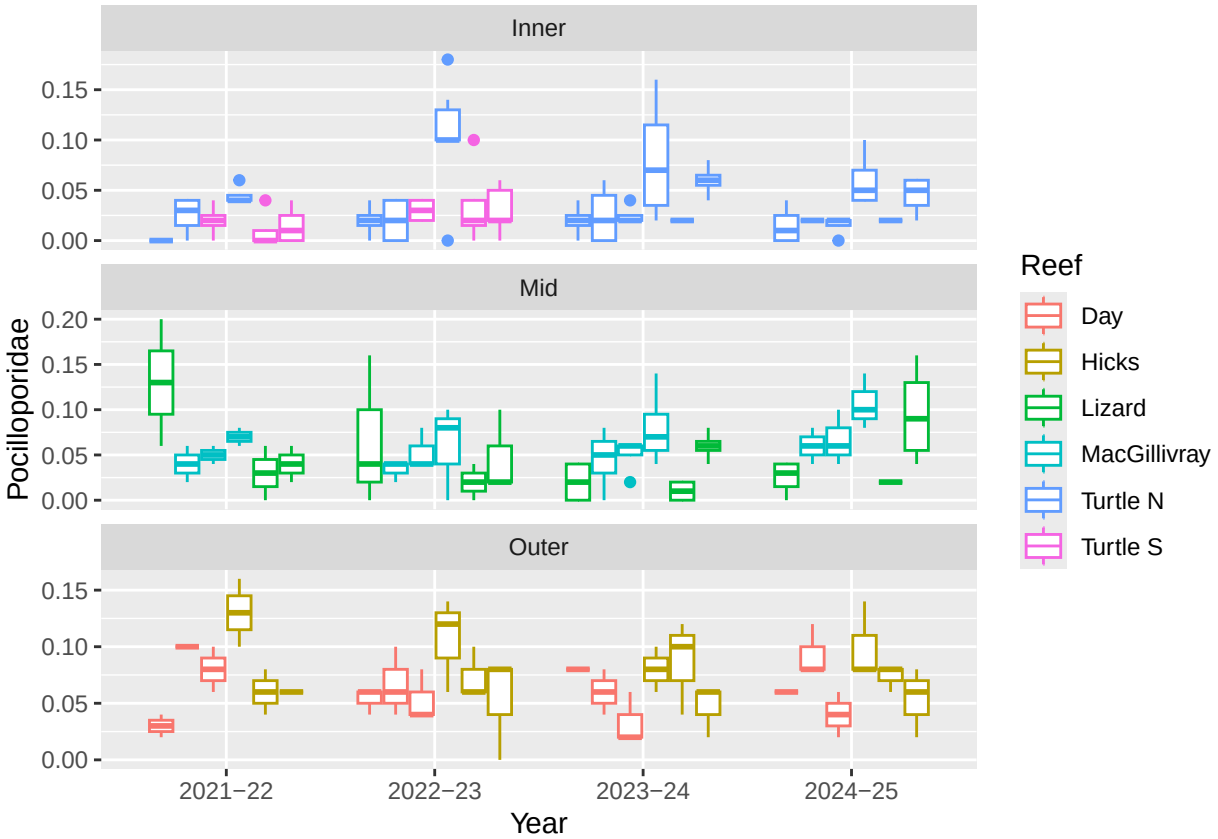
Acroporidae

```
ggplot(PIT_50m, aes(x=Year, y=Acroporidae, colour = Reef))+
  geom_boxplot(aes(group = interaction(Year, Site))) +
  facet_wrap(~Shelf, ncol = 1, scale = "free_y")
```



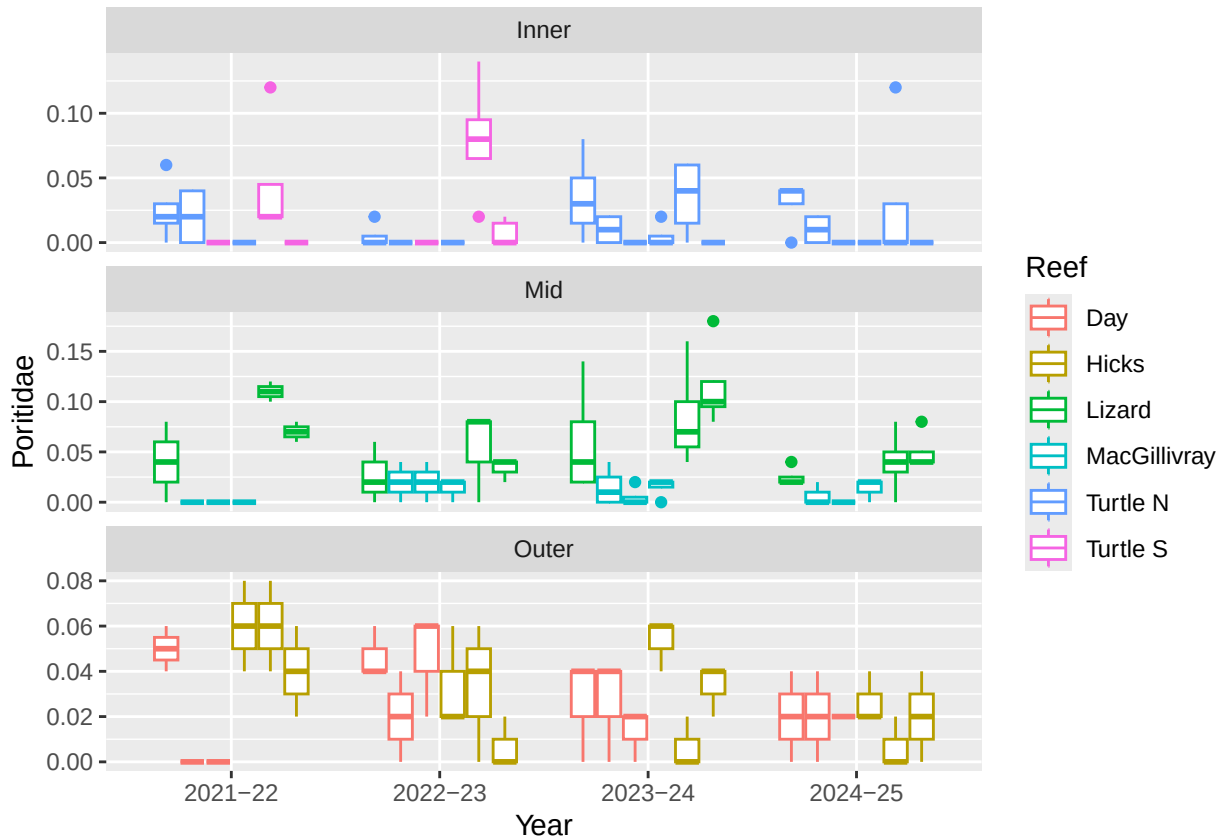
Pocilloporidae

```
ggplot(PIT_50m, aes(x=Year, y=Pocilloporidae, colour = Reef))+
  geom_boxplot(aes(group = interaction(Year, Site))) +
  facet_wrap(~Shelf, ncol = 1, scale = "free_y")
```



Poritidae

```
ggplot(PIT_50m, aes(x=Year, y=Poritidae, colour = Reef))+
  geom_boxplot(aes(group = interaction(Year, Site))) +
  facet_wrap(~Shelf, ncol = 1, scale = "free_y")
```



Why is there not benthic surveys at Turtle S in the past two years?

Victor Email: Regarding the PERMANOVA: you start with a community matrix (sites x coral taxa). A PERMANOVA tests whether community composition differs among groups, using a distance matrix derived from your multivariate raw data. The inputs are a dissimilarity (distance) matrix, and a grouping factor (i.e., a categorical variable you want to test, like "reef", "shelf.position", or "year"). In your case, I'd suggest calculating a Bray-Curtis distance matrix and then running a PERMANOVA with the vegan package. Here's a quick example you can run in R:

```
library(vegan)
abund <- PIT_50m |>
  select(Acroporidae, Pocilloporidae, Poritidae)

prop <- PIT_50m |>
  select(prop_coral, Prop_macroalgae)
```

Some of the rows do not have Acropora, Pocillopora or Porites present... should I then remove them? Because the distance matrix won't work otherwise.

```
metadata <- PIT_50m |>
  select(Site, Shelf, Reef, Year) |>
  mutate(
    Shelf = factor(Shelf),
    Reef = factor(Reef),
    Site = factor(Site)
  )
```

```r
PERMANOVA_prop <- adonis2(dist.mat ~ Reef:Shelf, data = metadata, permutations = 999, by = "terms")   #

print(PERMANOVA_prop)
```

```
## Permutation test for adonis under reduced model
## Terms added sequentially (first to last)
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = dist.mat ~ Reef:Shelf, data = metadata, permutations = 999, by = "terms")
##             Df SumOfSqs      R2     F Pr(>F)
## Reef:Shelf   5   21.173 0.58322 65.77  0.001 ***
## Residual   235   15.130 0.41678
## Total      240   36.303 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
PERMANOVA_prop2 <- adonis2(dist.mat ~ Year+Shelf+Year:Shelf, data = metadata, permutations = 999, by =

print(PERMANOVA_prop2)
```

```
## Permutation test for adonis under reduced model
## Terms added sequentially (first to last)
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = dist.mat ~ Year + Shelf + Year:Shelf, data = metadata, permutations = 999, by = "te
##             Df SumOfSqs      R2        F Pr(>F)
## Year         3    1.366 0.03763   7.2364  0.001 ***
## Shelf        2   18.956 0.52216 150.6158  0.001 ***
## Year:Shelf   6    1.570 0.04325   4.1587  0.001 ***
## Residual   229   14.410 0.39695
## Total      240   36.303 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

formula – model formula such as Y ~ A + B * C. See the 'Complex Models' chapter for more details of how to specify a model formula. For now, the key point is that we are specifying that we want to analyze the response (Y) as a function of (~) one or more explanatory variables (A, B, C). The class of Y determines what is done: If Y is a dissimilarity matrix (i.e., output from dist() or vegdist()), the pseudo-F statistic is calculated directly. If Y is a data frame or matrix containing data, vegdist() is applied to calculate the distance matrix first and then the pseudo-F statistic is calculated.

- 60% os total community variation (in dissimilarity matrix) explained by Year, SHelf, and their interaction. Shelf contributes 52.2% of total variation in community composition (proportion macroalgae and proportion coral)

###Try to do for abundances, but will need to remove rows where it is 0

```r
data_for_abundance <- PIT_50m |>
  mutate(clean = rowSums(across(c(Acroporidae, Pocilloporidae, Poritidae)), na.rm = TRUE))
```

```r
data_for_abundance <- data_for_abundance |>
  filter(clean > 0)


abund <- data_for_abundance |>
  select(Acroporidae, Pocilloporidae, Poritidae) |>
  mutate(
    Acroporidae = Acroporidae*50,
    Pocilloporidae = Pocilloporidae*50,
    Poritidae = Poritidae*50
  )

abund_std <- abund |>
  wisconsin()



abund_bray <- vegdist(abund_std, method = "bray")

meta_abund <- data_for_abundance |>
  select(Shelf, Year, Reef, Site)
```

```r
PERMANOVA_abund <- adonis2(abund_bray ~ Year+Shelf+ Year:Shelf, data = meta_abund, permutations = 999, b
```

```
## Permutation test for adonis under reduced model
## Terms added sequentially (first to last)
## Permutation: free
## Number of permutations: 999
##
## adonis2(formula = abund_bray ~ Year + Shelf + Year:Shelf, data = meta_abund, permutations = 999, by =
##             Df SumOfSqs      R2      F Pr(>F)
## Year         3   0.7571 0.02631 2.2329  0.045 *
## Shelf        2   1.1486 0.03992 5.0812  0.002 **
## Year:Shelf   6   1.4355 0.04990 2.1169  0.018 *
## Residual   225  25.4298 0.88387
## Total      236  28.7709 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Abundance is the community matrix data, with 237 sample units of 3 species.

```r
set.seed(42)
z <- metaMDS(comm = abund_std,
             autotransform = FALSE,
             distance = "bray",
             engine = "monoMDS",
             k=2,
             weakties = TRUE,
             model = "global",
             maxit = 300,
             try = 40,
             trymax = 100)
```

```
## Run 0 stress 0.03850562
## Run 1 stress 0.03850567
## ... Procrustes: rmse 2.913727e-05  max resid 0.0002612591
## ... Similar to previous best
## Run 2 stress 0.1023912
## Run 3 stress 0.03850574
## ... Procrustes: rmse 3.978818e-05  max resid 0.0003695499
## ... Similar to previous best
## Run 4 stress 0.03850568
## ... Procrustes: rmse 1.663054e-05  max resid 0.0002136687
## ... Similar to previous best
## Run 5 stress 0.07367531
## Run 6 stress 0.0729293
## Run 7 stress 0.03850563
## ... Procrustes: rmse 2.131298e-05  max resid 0.0002324154
## ... Similar to previous best
## Run 8 stress 0.0706665
## Run 9 stress 0.03850563
## ... Procrustes: rmse 2.048902e-05  max resid 0.0001848531
## ... Similar to previous best
## Run 10 stress 0.03850574
## ... Procrustes: rmse 3.717886e-05  max resid 0.0005056953
## ... Similar to previous best
## Run 11 stress 0.0385057
## ... Procrustes: rmse 3.410976e-05  max resid 0.0002980644
## ... Similar to previous best
## Run 12 stress 0.03850567
## ... Procrustes: rmse 1.359431e-05  max resid 0.0001518019
## ... Similar to previous best
## Run 13 stress 0.07681298
## Run 14 stress 0.146216
## Run 15 stress 0.03850563
## ... Procrustes: rmse 2.211932e-05  max resid 0.000193706
## ... Similar to previous best
## Run 16 stress 0.03850569
## ... Procrustes: rmse 3.20517e-05  max resid 0.0003779833
## ... Similar to previous best
## Run 17 stress 0.0736753
## Run 18 stress 0.03850563
## ... Procrustes: rmse 2.023263e-05  max resid 0.0002029496
## ... Similar to previous best
## Run 19 stress 0.03850566
## ... Procrustes: rmse 2.636962e-05  max resid 0.0002600825
## ... Similar to previous best
## Run 20 stress 0.03850562
## ... Procrustes: rmse 1.382034e-05  max resid 0.0001155798
## ... Similar to previous best
## Run 21 stress 0.07367527
## Run 22 stress 0.03850574
## ... Procrustes: rmse 3.836509e-05  max resid 0.0004505771
## ... Similar to previous best
## Run 23 stress 0.03850567
## ... Procrustes: rmse 1.318142e-05  max resid 0.0001422713
## ... Similar to previous best
```

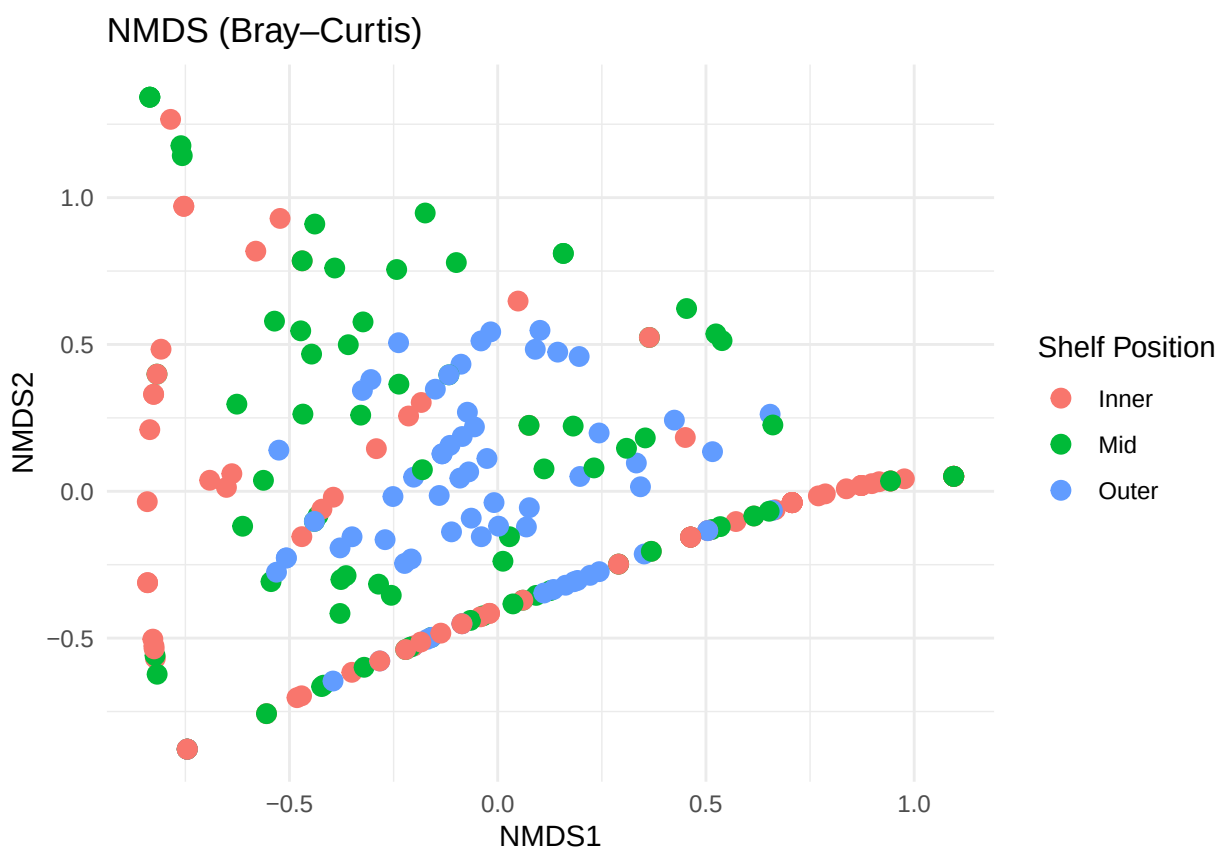
```
## Run 24 stress 0.03850573
## ... Procrustes: rmse 3.904847e-05  max resid 0.0003593751
## ... Similar to previous best
## Run 25 stress 0.03850562
## ... Procrustes: rmse 8.449854e-06  max resid 8.679783e-05
## ... Similar to previous best
## Run 26 stress 0.03850568
## ... Procrustes: rmse 1.358656e-05  max resid 0.0001417718
## ... Similar to previous best
## Run 27 stress 0.03850568
## ... Procrustes: rmse 3.188459e-05  max resid 0.0002949528
## ... Similar to previous best
## Run 28 stress 0.03850563
## ... Procrustes: rmse 1.938294e-05  max resid 0.0001819913
## ... Similar to previous best
## Run 29 stress 0.1122261
## Run 30 stress 0.03850572
## ... Procrustes: rmse 3.707697e-05  max resid 0.0004138189
## ... Similar to previous best
## Run 31 stress 0.07681296
## Run 32 stress 0.03850562
## ... Procrustes: rmse 1.979255e-05  max resid 0.0001720773
## ... Similar to previous best
## Run 33 stress 0.09480365
## Run 34 stress 0.03850573
## ... Procrustes: rmse 3.611839e-05  max resid 0.000453708
## ... Similar to previous best
## Run 35 stress 0.2133131
## Run 36 stress 0.03850569
## ... Procrustes: rmse 3.227999e-05  max resid 0.0003607917
## ... Similar to previous best
## Run 37 stress 0.07681297
## Run 38 stress 0.1426154
## Run 39 stress 0.03850574
## ... Procrustes: rmse 3.99838e-05  max resid 0.0004188733
## ... Similar to previous best
## Run 40 stress 0.03850564
## ... Procrustes: rmse 2.311891e-05  max resid 0.0002642151
## ... Similar to previous best
## *** Best solution repeated 26 times
```

```r
print(z)
```

```
##
## Call:
## metaMDS(comm = abund_std, distance = "bray", k = 2, try = 40,      trymax = 100, engine = "monoMDS",
##
## global Multidimensional Scaling using monoMDS
##
## Data:     abund_std
## Distance: bray
##
## Dimensions: 2
## Stress:     0.03850562
```

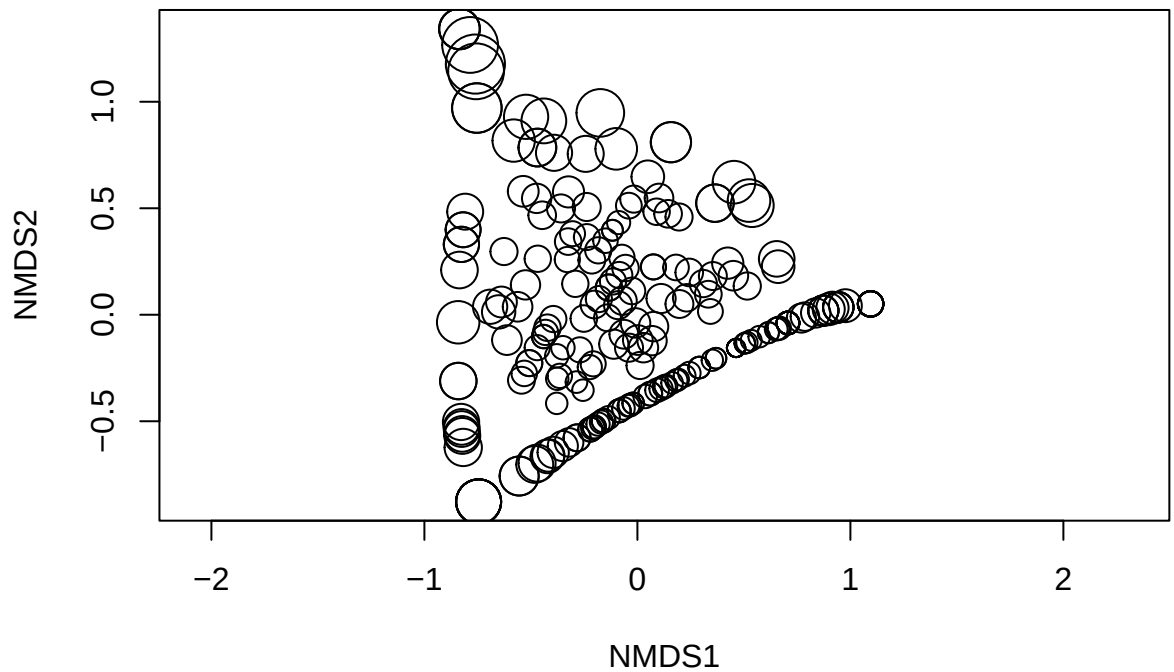
```
## Stress type 1, weak ties
## Best solution was repeated 26 times in 40 tries
## The best solution was from try 0 (metric scaling or null solution)
## Scaling: centring, PC rotation, halfchange scaling
## Species: expanded scores based on 'abund_std'
```

```r
# Extract NMDS site scores
scores_nmds <- as.data.frame(scores(z, display = "sites"))

# Join to metadata (assuming same row order)
scores_nmds <- scores_nmds |>
  bind_cols(meta_abund)


ggplot(scores_nmds, aes(x = NMDS1, y = NMDS2, colour = Shelf)) +
  geom_point(size = 3) +
  theme_minimal() +
  labs(
    title = "NMDS (Bray-Curtis)",
    colour = "Shelf Position"
  )
```



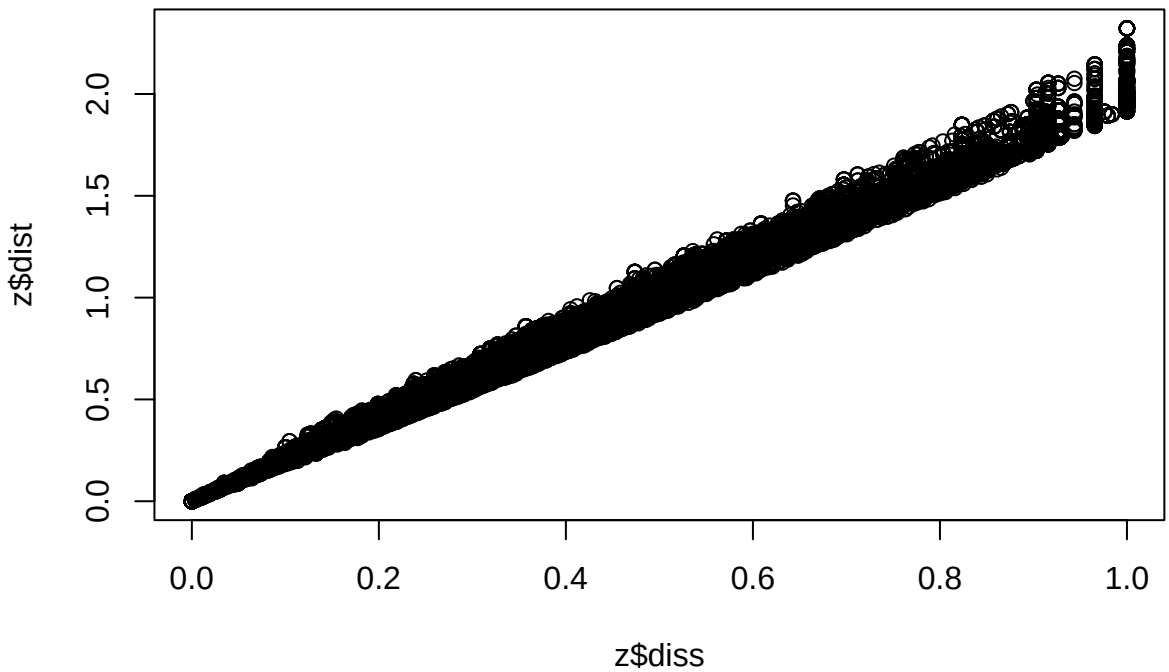
Stress <0.05 therefore excellent representation with no prospect of misinterpretation. https://uw.pressbooks.pub/appliedmultivariatestatistics/chapter/nmds/

```r
gof <- goodness(object = z)
plot(z, display = "sites", type = "none")
points(z, display = "sites", cex = 2*gof/mean(gof))
```

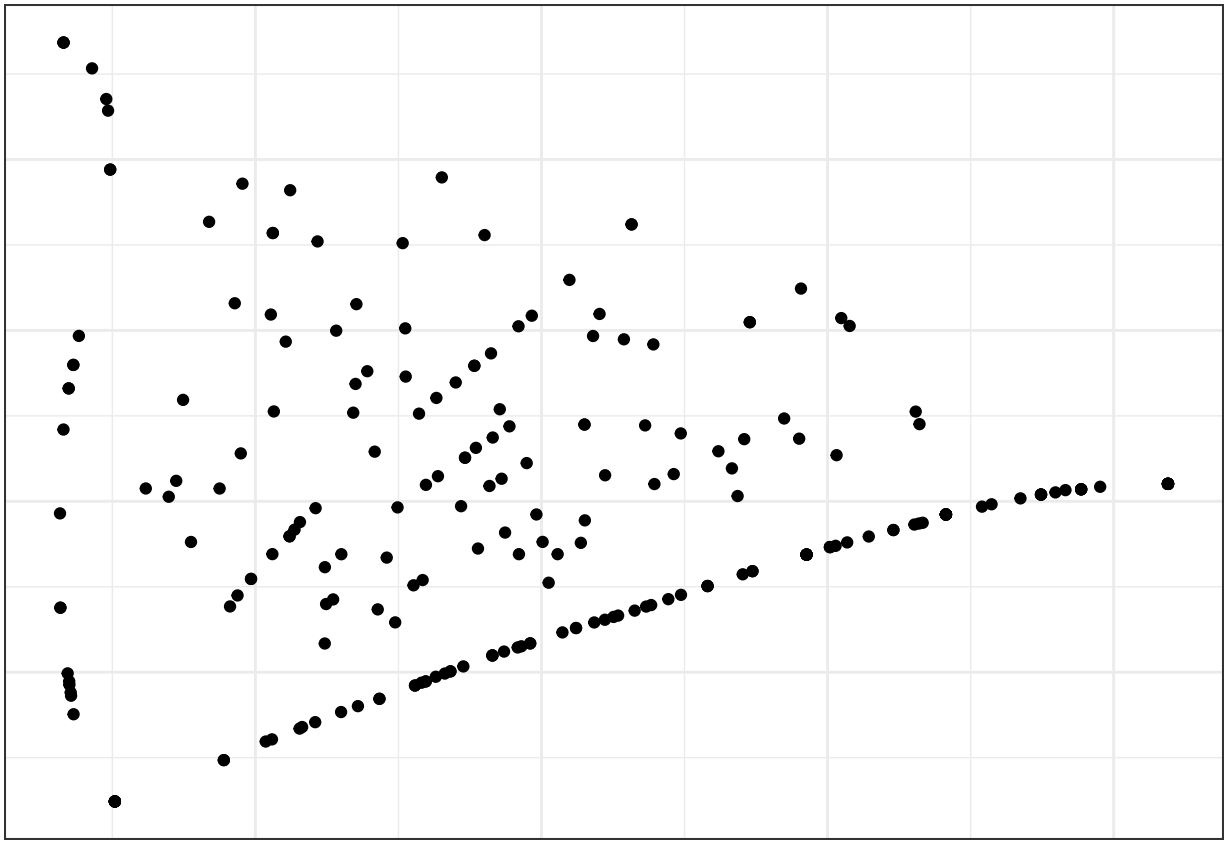


Evaluating fit plot original dissimilarities $(zdiss)against Euclidean ordination distances(z\text{dist})$.

```r
plot(z$diss, z$dist)
```




```
z.points <- data.frame(z$points,
                       Acroporidae = abund_std$Acroporidae,
                       Pocilloporidae = abund_std$Pocilloporidae,
                       Poritid = abund_std$Poritidae)

p <- ggplot(data = z.points, aes(x = MDS1, y = MDS2)) +
theme_bw() +
theme(axis.title = element_blank(),
  axis.ticks = element_blank(),
  axis.text = element_blank())

p + geom_point()
```



Correlation between brood stock and corals

Total hard coral cover and total recruits

```
PIT_50m |>
  group_by(Site, Year, Reef, Shelf) |>
  summarise(n = n())
```

```
## `summarise()` has grouped output by 'Site', 'Year', 'Reef'. You can override
## using the `.groups` argument.
```

```
## # A tibble: 72 x 5
## # Groups:   Site, Year, Reef [72]
##    Site  Year    Reef  Shelf     n
##    <fct> <chr>   <fct> <fct> <int>
##  1 Day 1 2021-22 Day   Outer     2
##  2 Day 1 2022-23 Day   Outer     3
##  3 Day 1 2023-24 Day   Outer     3
##  4 Day 1 2024-25 Day   Outer     3
##  5 Day 2 2021-22 Day   Outer     2
##  6 Day 2 2022-23 Day   Outer     3
##  7 Day 2 2023-24 Day   Outer     3
```

```
##  8 Day 2 2024-25 Day   Outer     3
##  9 Day 3 2021-22 Day   Outer     2
## 10 Day 3 2022-23 Day   Outer     3
## # i 62 more rows
```

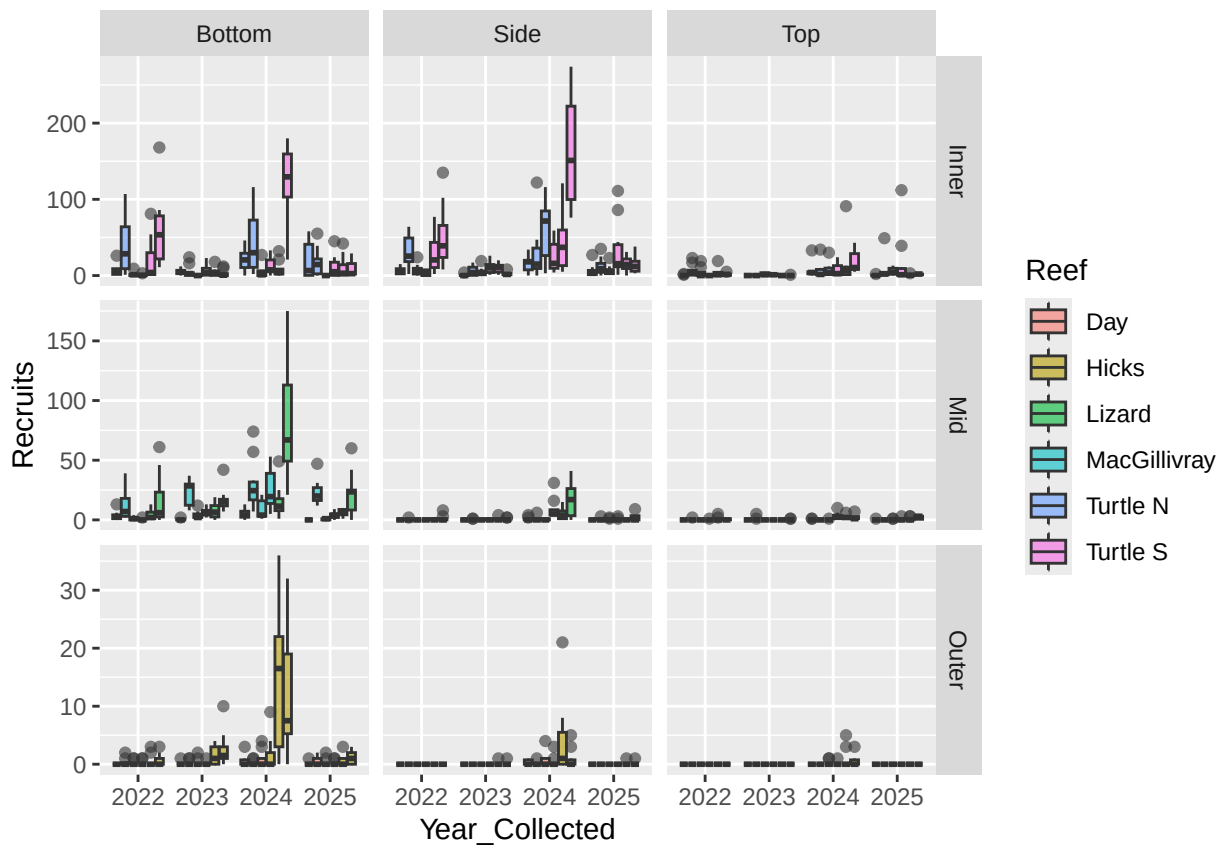
Would need to do average or something to have a single value for PIT and single value for tiles for a given combination of site, year, and recruits/brood stock.

Total Acroporidae cover and total acroproidae recruits

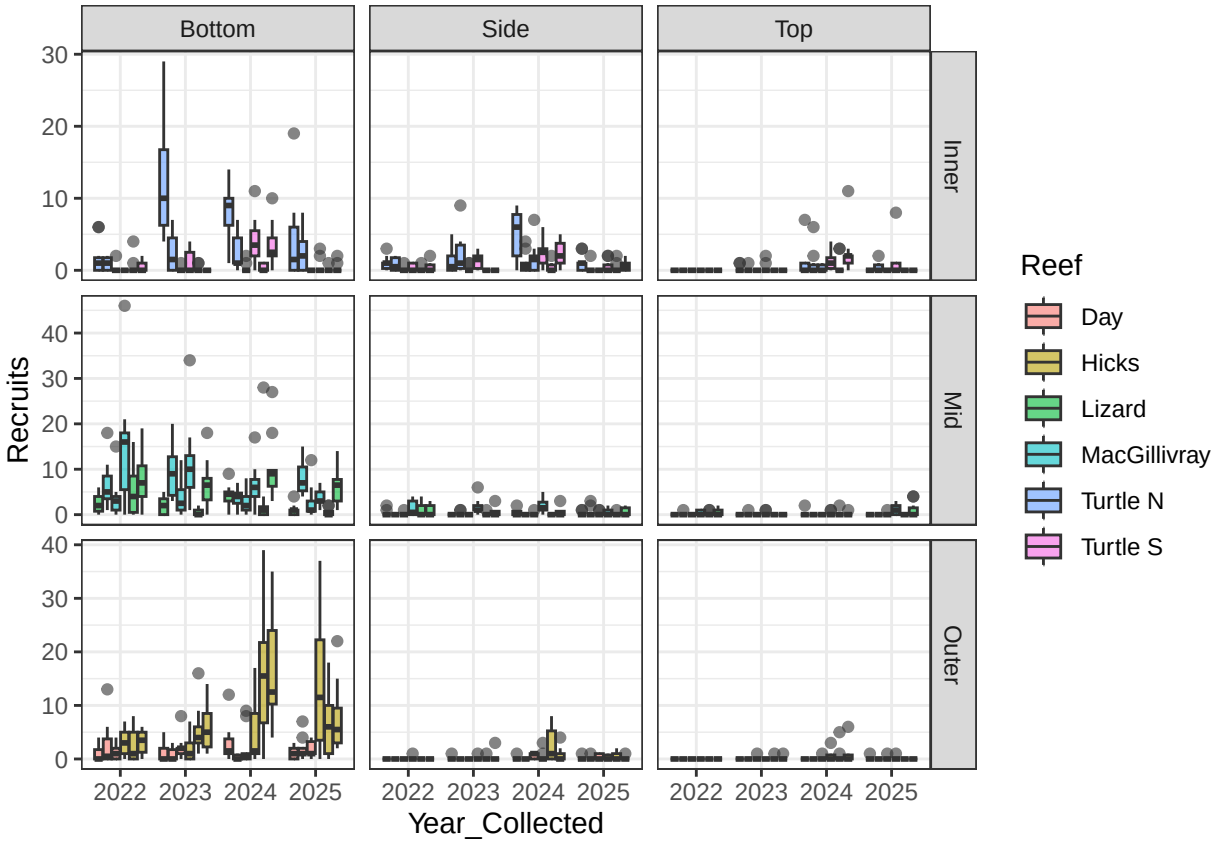
Total Pocilloporidae cover and total Pocillopordiae recruits

Position on tile (IGNORE)

Acroporidae

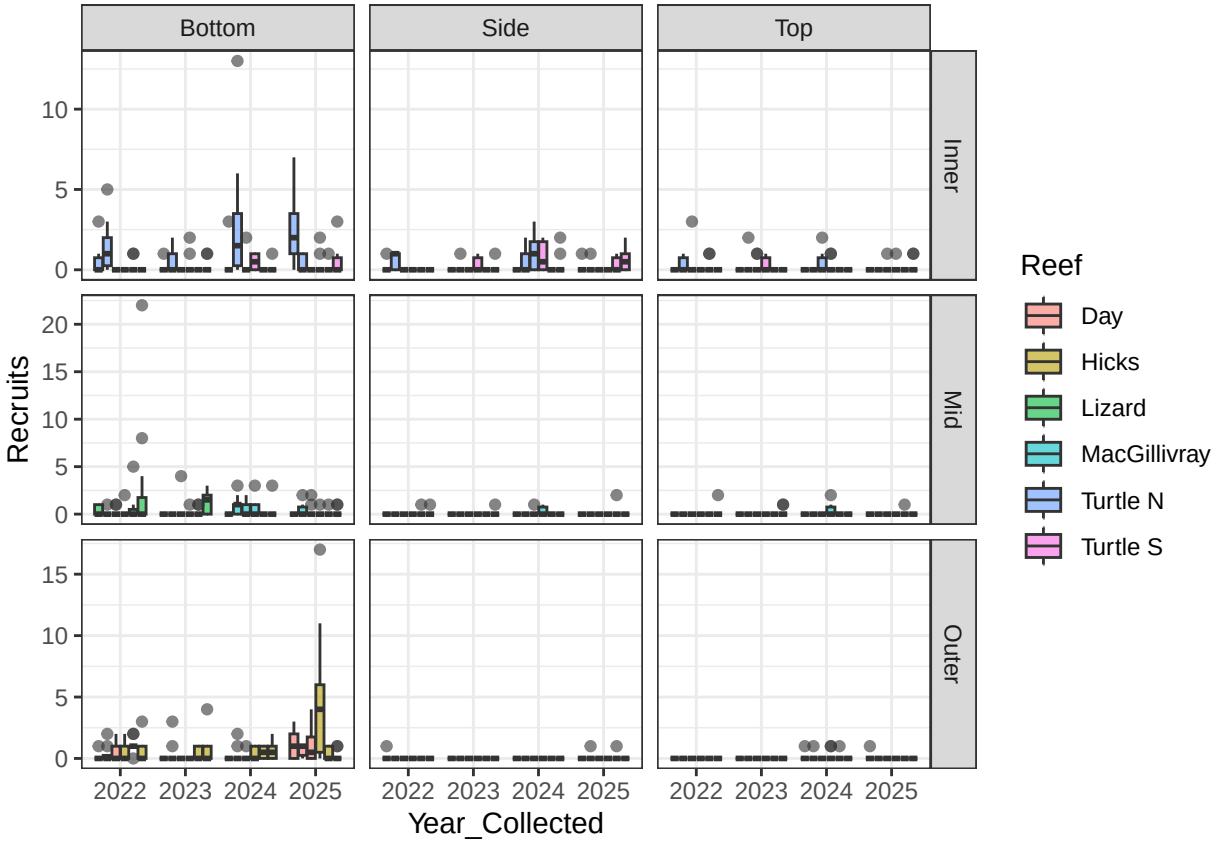


Pocilloporidae



Poritidae

```
Poritid_orientation |>
  ggplot(aes(x = Year_Collected, y = Recruits, fill = Reef)) +
  geom_boxplot(aes(group = interaction(Year_Collected, Site)),
               position = position_dodge(width = 0.8),
               alpha = 0.6) +
  facet_grid(Shelf_Position~Tile_Surface, scale = "free_y")+
  theme_bw()
```


Bottom
Side
Top
Inner
Mid
Outer
Recruits
Year_Collected
2022
2023
2024
2025
Reef
Day
Hicks
Lizard
MacGillivray
Turtle N
Turtle S